Lab HA/BCP — Building a Proxmox cluster and testing it under failure

GRETA de l'Essonne — AIS professional title — Block 2: Design and evolution

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Charles Maduba — Group 3 (work carried out in a team of 4 trainees)

Session A: Monday 5 October 2026 — Parts 1 and 2. Session B: Thursday 8 October 2026 — Parts 3 and 4.

English version of the lab report, for personal reference. The original report was written in French.

Summary

Part	Content	Status
1	Prepare 3 Proxmox servers (pve1, pve2, pve3)	Done
2	Create the cluster and check quorum	Done
3	Shared storage (NFS)	Done
4	HA, failure test, RTO measurement	Done (RTO: 2 min 17 s)

Node	Name	IP address
Node 1 (existing)	pve (alias pve1)	192.168.56.200
Node 2	pve2	192.168.56.201
Node 3	pve3	192.168.56.202

PART 1 — Preparing the 3 servers

1.1 — Creating pve2 and pve3

Note: pve2 and pve3 were not cloned from pve1. They were installed from scratch from the Proxmox VE ISO, in VirtualBox, on the same host-only network 192.168.56.0/24. There are therefore no screenshots of the clone wizard.

Consequence: each node got its own SSH keys and its own machine identifier (machine-id) at installation. Step 1.3 (regenerating keys and machine-id) was not needed; we only checked that the three identifiers are different.

1.2 — Name, /etc/hosts file and IP address of each node

```
root@pve:~# cat /etc/hosts
127.0.0.1 localhost.localdomain localhost
192.168.56.200 pve.local.lan pve pve1
192.168.56.201 pve2.lab.local pve2
192.168.56.202 pve3.lab.local pve3

# The following lines are desirable for IPv6 capable hosts

::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
root@pve:~# hostname -i
192.168.56.200
root@pve:~# cat /etc/machine-id
f4679ee0fbab48b2bebf6f1afdb5e526
root@pve:~#
```

pve1: /etc/hosts with the 3 nodes, hostname -i = 192.168.56.200, machine-id

```
root@pve2:~# cat /etc/hosts
127.0.0.1 localhost.localdomain localhost
192.168.56.201 pve2.lab.local pve2
192.168.56.200 pve.local.lan pve pve1
192.168.56.202 pve3.lab.local pve3

# The following lines are desirable for IPv6 capable hosts

::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
root@pve2:~# hostname -i
192.168.56.201
root@pve2:~# ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP
enp0s8           UP             10.0.3.15/24 fd17:625c:f037:3:a00:27ff:feef:ef90/64 fe80::a00:27ff:feef:ef90/64
vmbr0            UP             192.168.56.201/24 fe80::a00:27ff:fef6:a204/64
root@pve2:~#
```

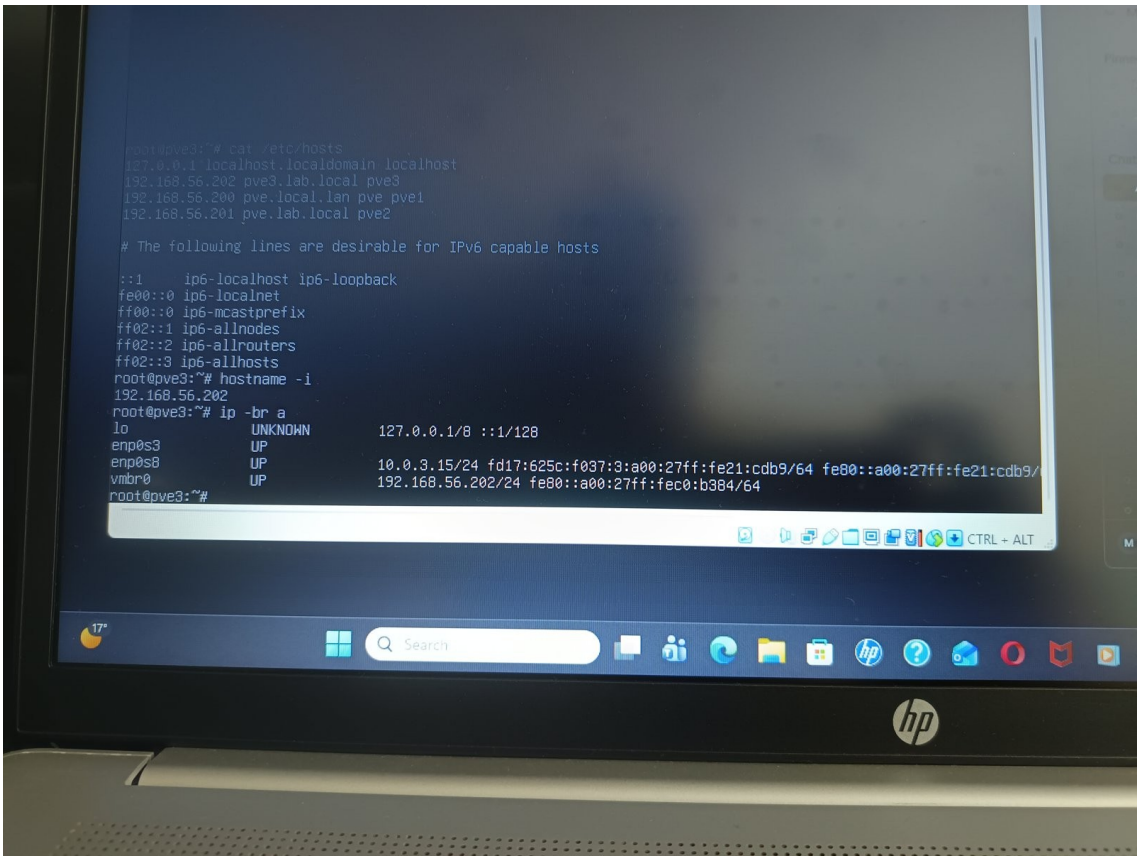
pve2: /etc/hosts, hostname -i and ip -br a (vmbr0 = 192.168.56.201)

```
root@pve3:~# sed -i 's/201 pve.lab.local/201 pve2.lab.local/' /etc/hosts
root@pve3:~# cat /etc/hosts
127.0.0.1 localhost.localdomain localhost
192.168.56.202 pve3.lab.local pve3
192.168.56.200 pve.local.lan pve pve1
192.168.56.201 pve2.lab.local pve2

# The following lines are desirable for IPv6 capable hosts

::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
root@pve3:~#
```

pve3: contents of /etc/hosts



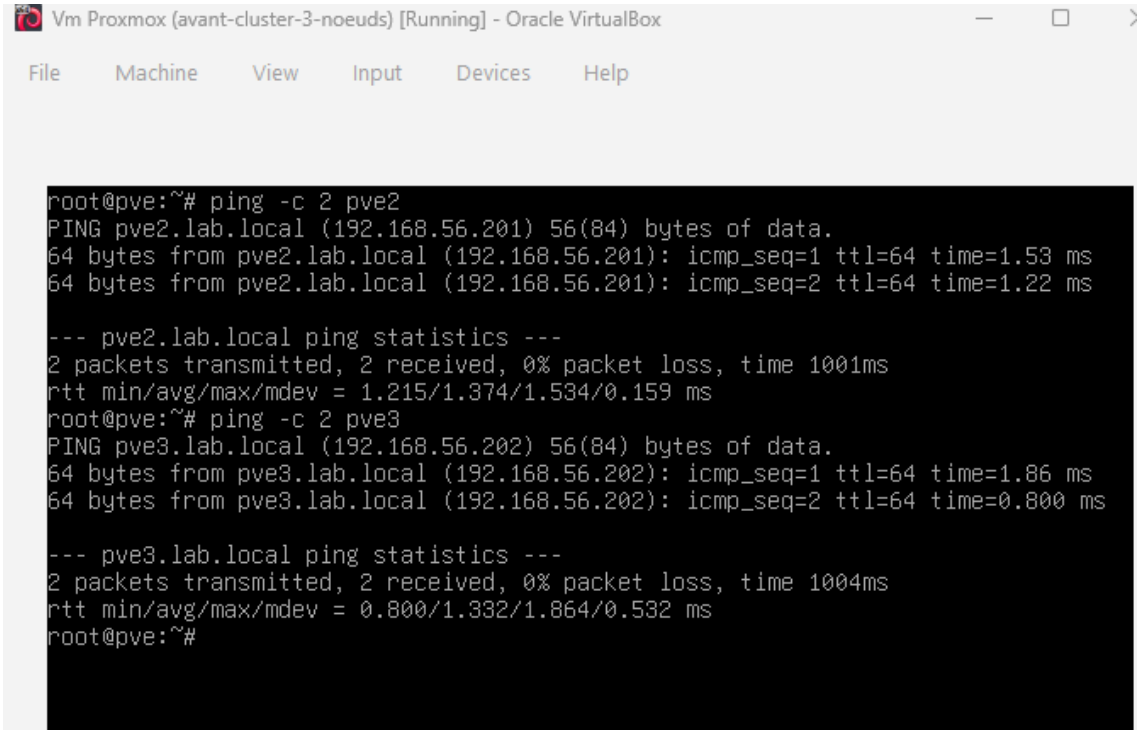
pve3: hostname -i and ip -br a (vmbr0 = 192.168.56.202)

1.3 — The 3 machine-ids

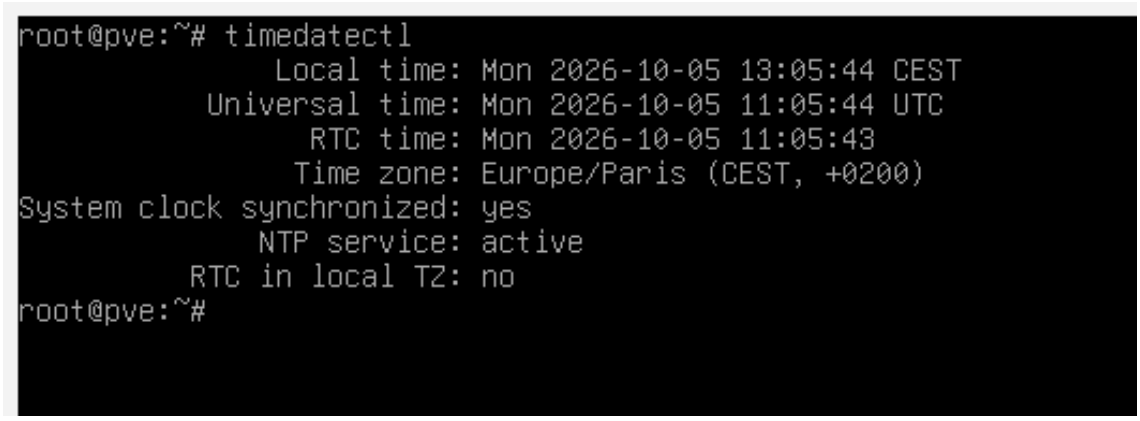
Node	machine-id
pve1	f4679ee0fbab48b2bebf6f1afdb5e526
pve2	0c5c5b5745d04cf2985687e8c53da75f
pve3	c895ad2836d5419fb7c48d0328d078e3

The three identifiers are different (screenshots above for pve1, below for pve2 and pve3).

1.4 — The 3 nodes can see each other (ping, time, empty nodes)



pve1: ping -c 2 pve2 and ping -c 2 pve3 — each name answers with the right IP address



pve1: timedatectl — System clock synchronized: yes

```
root@pve2:~# cat /etc/machine-id
0c5c5b5745d04cf2985687e8c53da75f
root@pve2:~# ping -c 2 pve1
PING pve.local.lan (192.168.56.200) 56(84) bytes of data.
64 bytes from pve.local.lan (192.168.56.200): icmp_seq=1 ttl=64 time=1.33 ms
64 bytes from pve.local.lan (192.168.56.200): icmp_seq=2 ttl=64 time=0.620 ms

--- pve.local.lan ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 0.620/0.974/1.329/0.354 ms
root@pve2:~# ping -c 2 pve3
PING pve3.lab.local (192.168.56.202) 56(84) bytes of data.
64 bytes from pve3.lab.local (192.168.56.202): icmp_seq=1 ttl=64 time=5.83 ms
64 bytes from pve3.lab.local (192.168.56.202): icmp_seq=2 ttl=64 time=0.584 ms

--- pve3.lab.local ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1107ms
rtt min/avg/max/mdev = 0.584/3.207/5.831/2.623 ms
root@pve2:~# timedatectl
               Local time: Mon 2026-10-05 12:42:18 CEST
           Universal time: Mon 2026-10-05 10:42:18 UTC
                 RTC time: Mon 2026-10-05 10:42:18
                Time zone: Europe/Paris (CEST, +0200)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
root@pve2:~# qm list
root@pve2:~# pct list
root@pve2:~#
```

pve2: machine-id, ping pve1 and pve3, timedatectl (synchronised), qm list and pct list empty

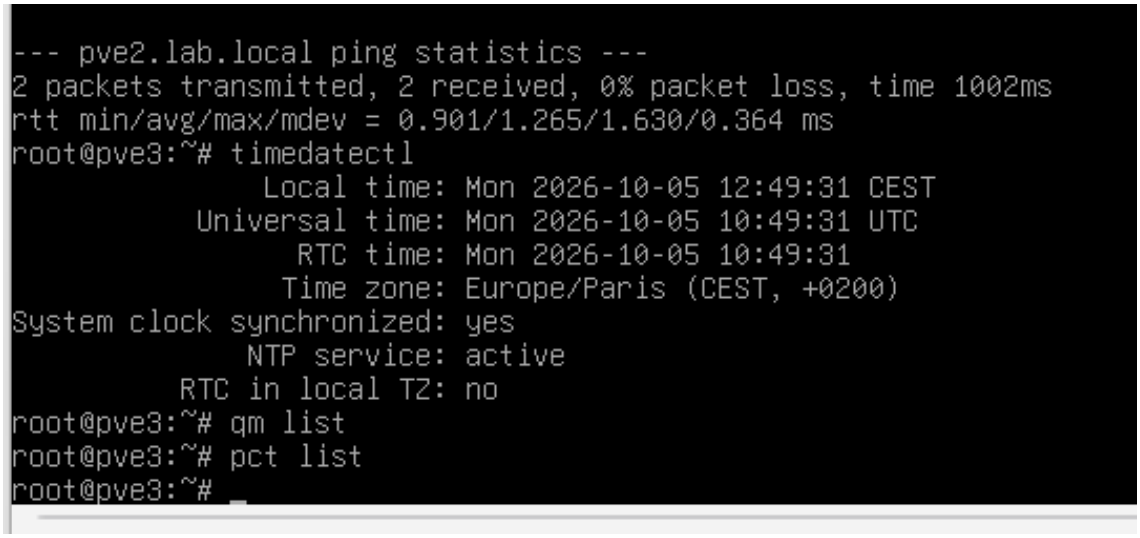
```
192.168.56.202
root@pve3:~# ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP
enp0s8           UP             10.0.3.15/24 fd17:625c:f037:3:a00:27ff:fe21:cdb9/64 fe80::a00:27ff:fe21:cdb9
vmbr0            UP             192.168.56.202/24 fe80::a00:27ff:fec0:b384/64
root@pve3:~# sed -i 's/201 pve.lab.local/201 pve2.lab.local/' /etc/hosts
root@pve3:~# cat /etc/hosts
127.0.0.1 localhost.localdomain localhost
192.168.56.202 pve3.lab.local pve3
192.168.56.200 pve.local.lan pve pve1
192.168.56.201 pve2.lab.local pve2

# The following lines are desirable for IPv6 capable hosts

::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
root@pve3:~# cat /etc/machine-id
c895ad2836d5419fb7c48d0328d078e3
root@pve3:~# ping -c 2 pve1
PING pve.local.lan (192.168.56.200) 56(84) bytes of data.
64 bytes from pve.local.lan (192.168.56.200): icmp_seq=1 ttl=64 time=2.37 ms
64 bytes from pve.local.lan (192.168.56.200): icmp_seq=2 ttl=64 time=0.647 ms

--- pve.local.lan ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.647/1.508/2.370/0.861 ms
root@pve3:~# ping -c 2 pve2
PING pve2.lab.local (192.168.56.201) 56(84) bytes of data.
64 bytes from pve2.lab.local (192.168.56.201): icmp_seq=1 ttl=64 time=1.63 ms
64 bytes from pve2.lab.local (192.168.56.201): icmp_seq=2 ttl=64 time=0.901 ms
```

pve3: ip -br a, /etc/hosts, machine-id, ping pve1 and pve2

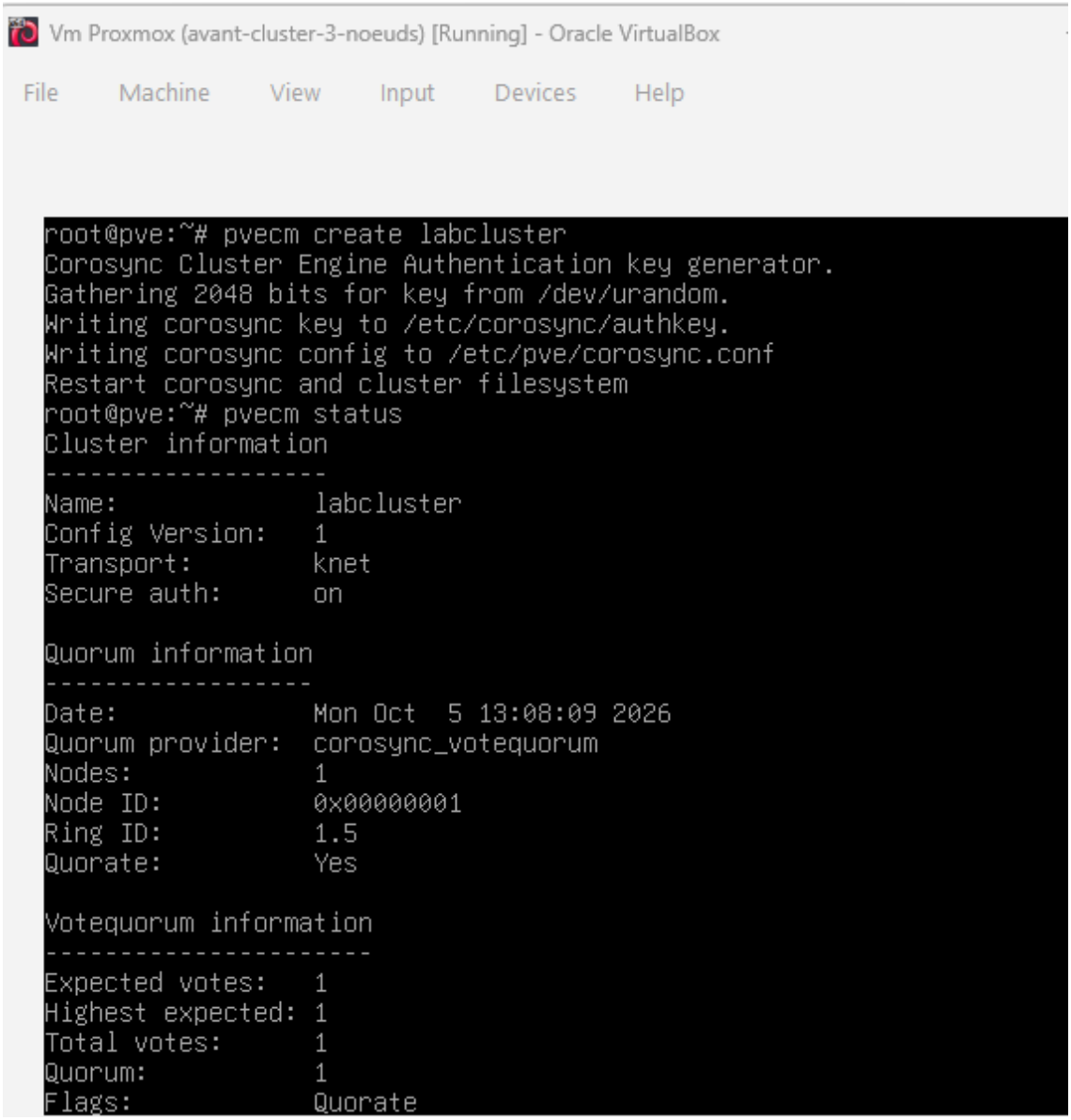


pve3: timedatectl (synchronised), qm list and pct list empty

Result: each name answers with the right IP address, the clock is synchronised on all 3 nodes, and pve2 and pve3 contain no VMs or containers (a requirement to join a cluster).

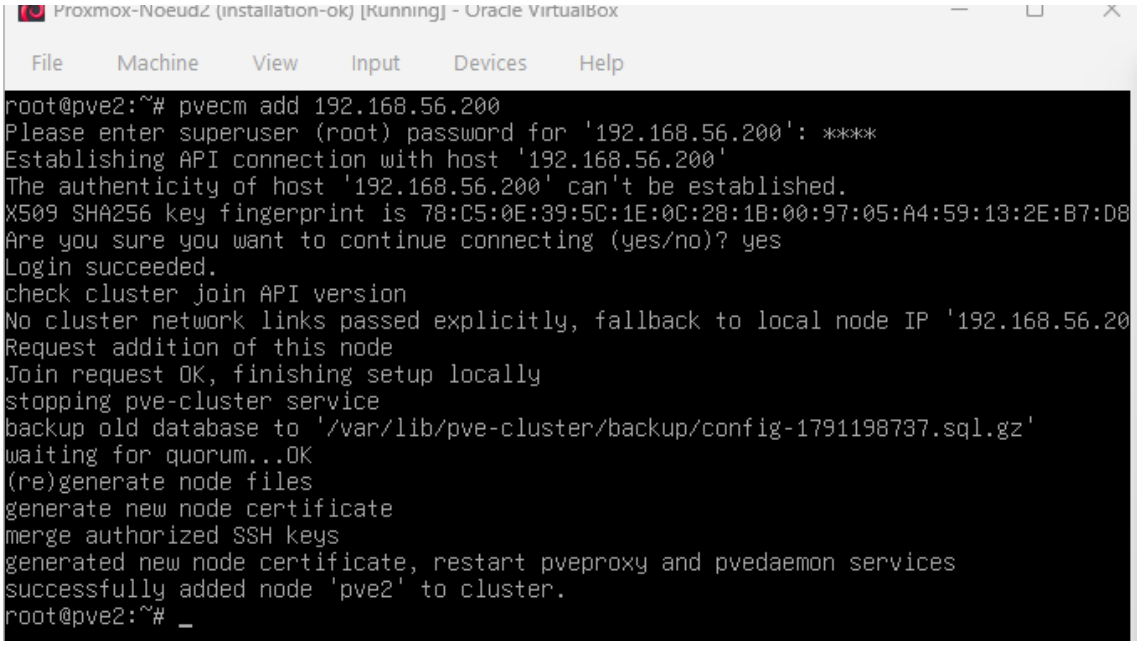
PART 2 — Creating the cluster

2.1 / 2.2 — Creating the cluster on pve1 and first pvecm status



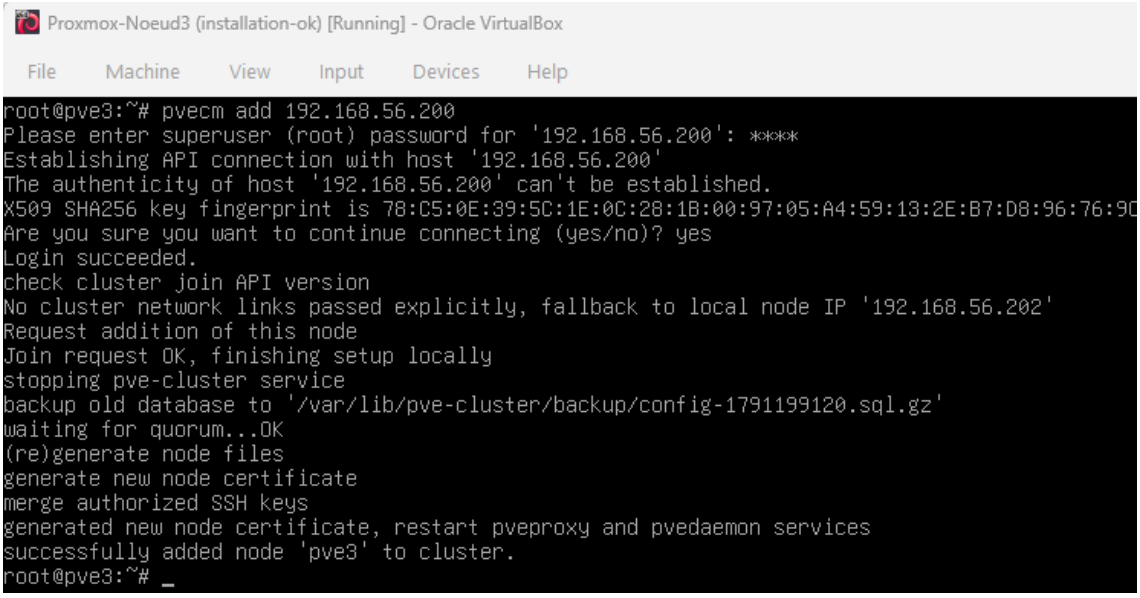
pvecm create labcluster then pvecm status: Nodes: 1, Quorate: Yes

2.3 — Adding pve2



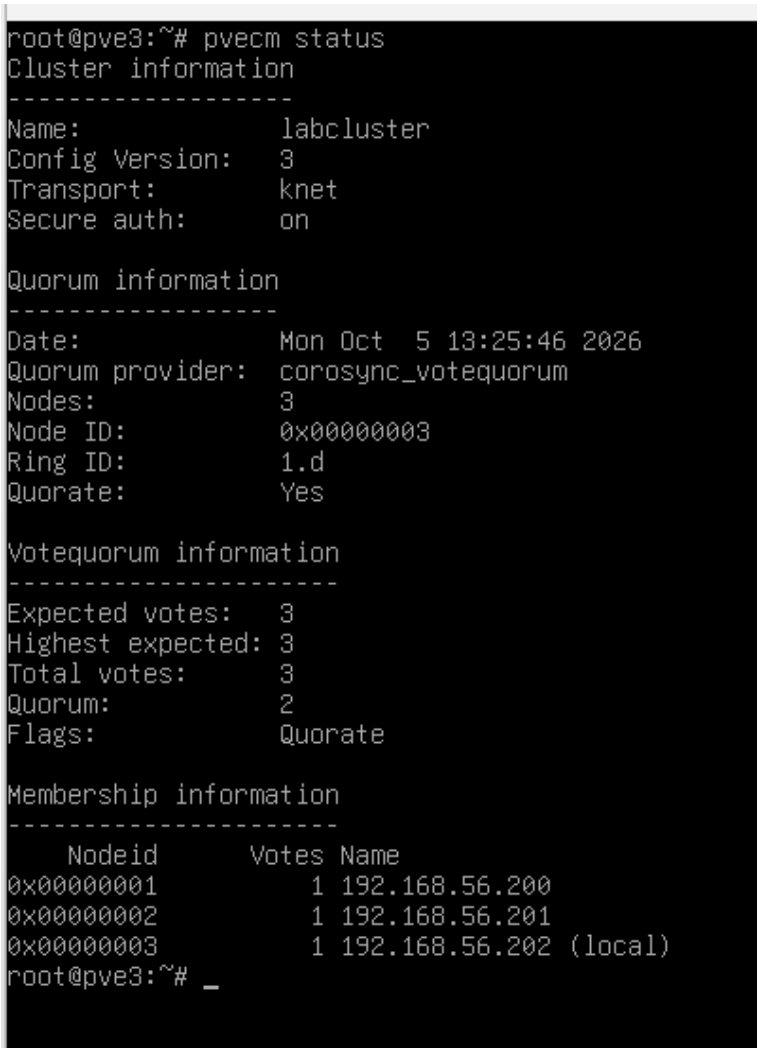
pve2: pvecm add 192.168.56.200 — node added to the cluster

2.4 — Adding pve3



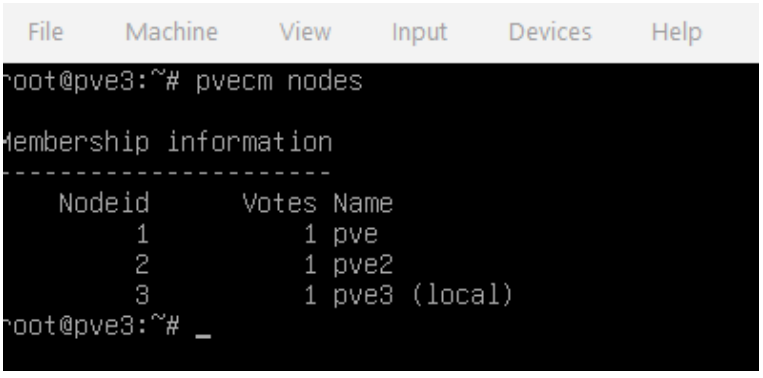
pve3: pvecm add 192.168.56.200 — node added to the cluster

2.5 — pvecm status with 3 nodes



Nodes: 3, Total votes: 3, Quorum: 2, Quorate: Yes

2.6 — pvecm nodes

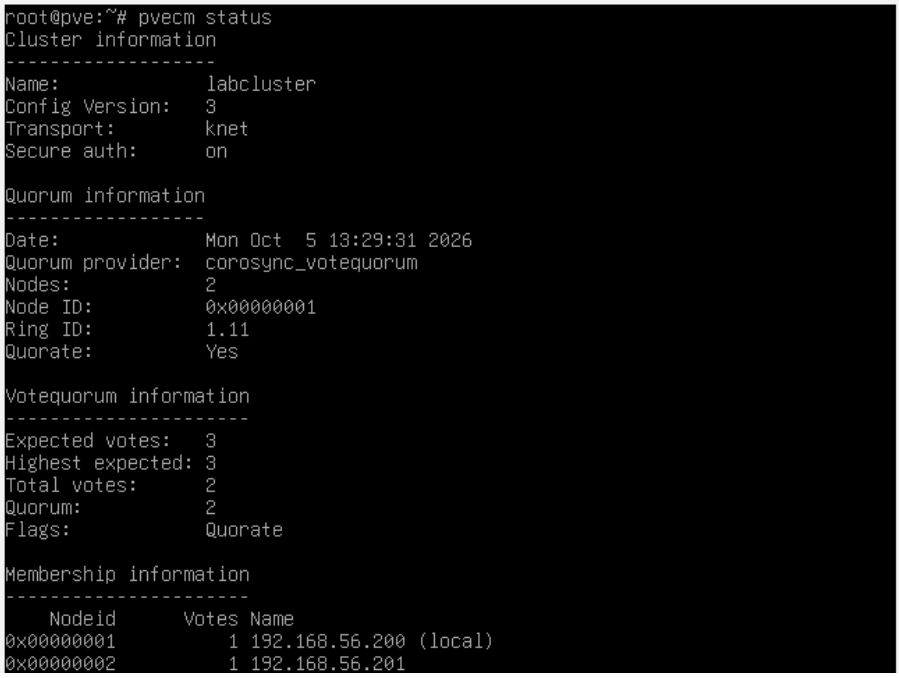


pve, pve2 and pve3 listed

2.7 — Experiment: quorum live

Situation	Votes present	Quorate	Observation

3 nodes on	3 / 3	Yes	Normal cluster
pve3 off	2 / 3	Yes	The cluster keeps working
pve2 and pve3 off	1 / 3	No	"Activity blocked": the cluster refuses to act
pve2 and pve3 back on	3 / 3	Yes	Back to normal after 1 to 2 minutes



pve3 off: Nodes: 2, Quorate: Yes

```
root@pve:~# pvecm status
Cluster information
-------------------
Name:             labcluster
Config Version:   3
Transport:        knet
Secure auth:      on

Quorum information
------------------
Date:             Mon Oct  5 13:32:30 2026
Quorum provider:  corosync_votequorum
Nodes:            1
Node ID:          0x00000001
Ring ID:          1.15
Quorate:          No

Votequorum information
----------------------
Expected votes:   3
Highest expected: 3
Total votes:      1
Quorum:           2 Activity blocked
Flags:

Membership information
----------------------
    Nodeid      Votes Name
0x00000001          1 192.168.56.200 (local)
root@pve:~# _
```

pve2 and pve3 off: Nodes: 1, Quorate: No, "Activity blocked"

```
root@pve:~# pvecm status
Cluster information
-------------------
Name:             labcluster
Config Version:   3
Transport:        knet
Secure auth:      on

Quorum information
------------------
Date:             Mon Oct  5 13:44:12 2026
Quorum provider:  corosync_votequorum
Nodes:            3
Node ID:          0x00000001
Ring ID:          1.1e
Quorate:          Yes

Votequorum information
----------------------
Expected votes:   3
Highest expected: 3
Total votes:      3
Quorum:           2
Flags:            Quorate

Membership information
----------------------
    Nodeid      Votes Name
0x00000001          1 192.168.56.200 (local)
0x00000002          1 192.168.56.201
```

pve2 and pve3 back on: Nodes: 3, Quorate: Yes

Question: why are 2 nodes out of 3 enough, while 1 node out of 3 is not?

Quorum is the majority of votes: quorum = (number of nodes ÷ 2) rounded down + 1. With 3 nodes: 3 ÷ 2 = 1.5 → 1, then 1 + 1 = 2 votes.

With 2 nodes on, there are 2 votes ≥ 2: the cluster has quorum and keeps working.

With only 1 node, there is 1 vote < 2: no quorum, the cluster blocks actions ("Activity blocked"). This is the protection against split-brain: an isolated node must not act alone, because it cannot know whether the others are still working on their side. If it did, two parts of the cluster could start the same VM at the same time and damage its data.

PART 3 — Shared storage (NFS)

Done on the command line (pvesm / qm) instead of the web interface: with the 3 nodes running, the host PC did not have enough memory to open a browser. The result is identical; the web interface uses the same commands.

3.1 — Installing the NFS server on pve and creating the shared folder

Commands: apt install -y nfs-kernel-server then mkdir -p /srv/nfs/vmstore. Before installing, we checked pve's internet access (ping deb.debian.org: 0% loss).

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  nfs-kernel-server
0 upgraded, 1 newly installed, 0 to remove and 149 not upgraded.
Need to get 152 kB of archives.
After this operation, 626 kB of additional disk space will be used.
Get:1 http://ftp.fr.debian.org/debian bookworm/main amd64 nfs-kernel-server amd6
4 1:2.6.2-4+deb12u1 [152 kB]
Fetched 152 kB in 0s (1,569 kB/s)
Selecting previously unselected package nfs-kernel-server.
(Reading database ... 46310 files and directories currently installed.)
Preparing to unpack .../nfs-kernel-server_1%3a2.6.2-4+deb12u1_amd64.deb ...
Unpacking nfs-kernel-server (1:2.6.2-4+deb12u1) ...
Setting up nfs-kernel-server (1:2.6.2-4+deb12u1) ...
Created symlink /etc/systemd/system/nfs-client.target.wants/nfs-blkmap.service →
 /lib/systemd/system/nfs-blkmap.service.
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service →
 /lib/systemd/system/nfs-server.service.
nfs-mountd.service is a disabled or a static unit, not starting it.
nfsdcld.service is a disabled or a static unit, not starting it.

Creating config file /etc/exports with new version

Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for man-db (2.11.2-2) ...
root@pve:~# mkdir -p /srv/nfs/vmstore
root@pve:~#
```

nfs-kernel-server installed without errors, /srv/nfs/vmstore created

3.2 — Declaring the share (exportfs -v)

Line added to /etc/exports: /srv/nfs/vmstore 192.168.56.0/24(rw,sync,no_subtree_check,no_root_squash), then exportfs -ra. Only the lab network (192.168.56.0/24) can access the share.

```
root@pve:~# mkdir -p /srv/nfs/vmstore
root@pve:~# echo '/srv/nfs/vmstore 192.168.56.0/24(rw,sync,no_subtree_check,no_r
oot_squash)' >> /etc/exports
root@pve:~# cat /etc/exports
# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subt
ree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/srv/nfs/vmstore 192.168.56.0/24(rw,sync,no_subtree_check,no_root_squash)
root@pve:~# exportfs-ra
-bash: exportfs-ra: command not found
root@pve:~# exportfs -ra
root@pve:~# exportfs -v
/srv/nfs/vmstore
                192.168.56.0/24(sync,wdelay,hide,no_subtree_check,sec=sys,rw,sec
ure,no_root_squash,no_all_squash)
root@pve:~#
```

cat /etc/exports and exportfs -v: the /srv/nfs/vmstore share is active

3.3 — nfs-shared storage added to the cluster (visible on the 3 nodes)

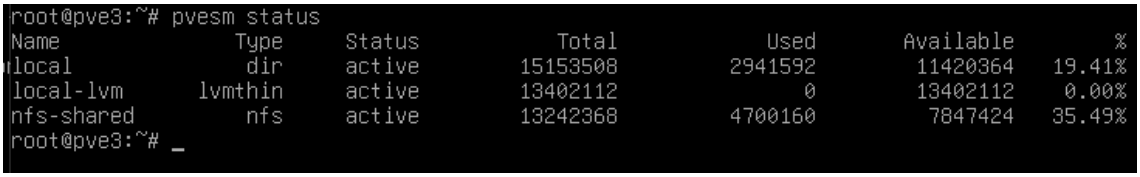
Command on pve: pvesm add nfs nfs-shared --server 192.168.56.200 --export /srv/nfs/vmstore --content images. The storage is defined at Datacenter level: pve2 and pve3 receive it automatically.

```
root@pve:~# systemctl restart pve-cluster
root@pve:~# pvesm add nfs nfs-shared --server 192.168.56.200 --export /srv/nfs/v
mstore --content images
root@pve:~# pvesm status
Name             Type     Status           Total            Used       Availabl
e        %
local             dir     active        13242248         4698152         784962
0   35.48%
local-lvm     lvmthin     active        11460608         9796527         166408
0   85.48%
nfs-shared        nfs     active        13242368         4698112         784998
4   35.48%
root@pve:~#
```

pve: nfs-shared, type nfs, status active

```
root@pve2:~# pvesm status
Name             Type     Status           Total            Used       Available        %
local             dir     active        15153508         2950036        11411920   19.47%
local-lvm     lvmthin     active        13402112               0        13402112    0.00%
nfs-shared        nfs     active        13242368         4700160         7847424   35.49%
root@pve2:~#
```

pve2: nfs-shared active

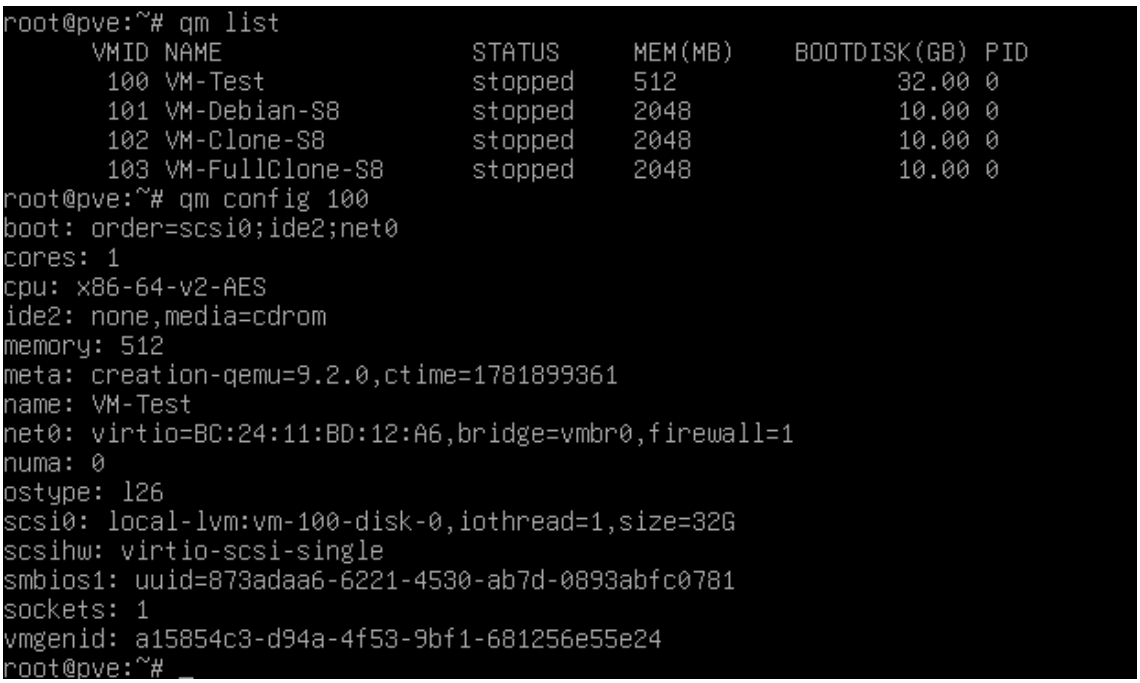


pve3: nfs-shared active

Note: a first attempt failed with "cfs-lock … error: no quorum" while pvecm status showed Quorate: Yes. The cluster configuration service (pmxcfs) had stayed read-only after the nodes briefly lost contact. Fixed with systemctl restart pve-cluster.

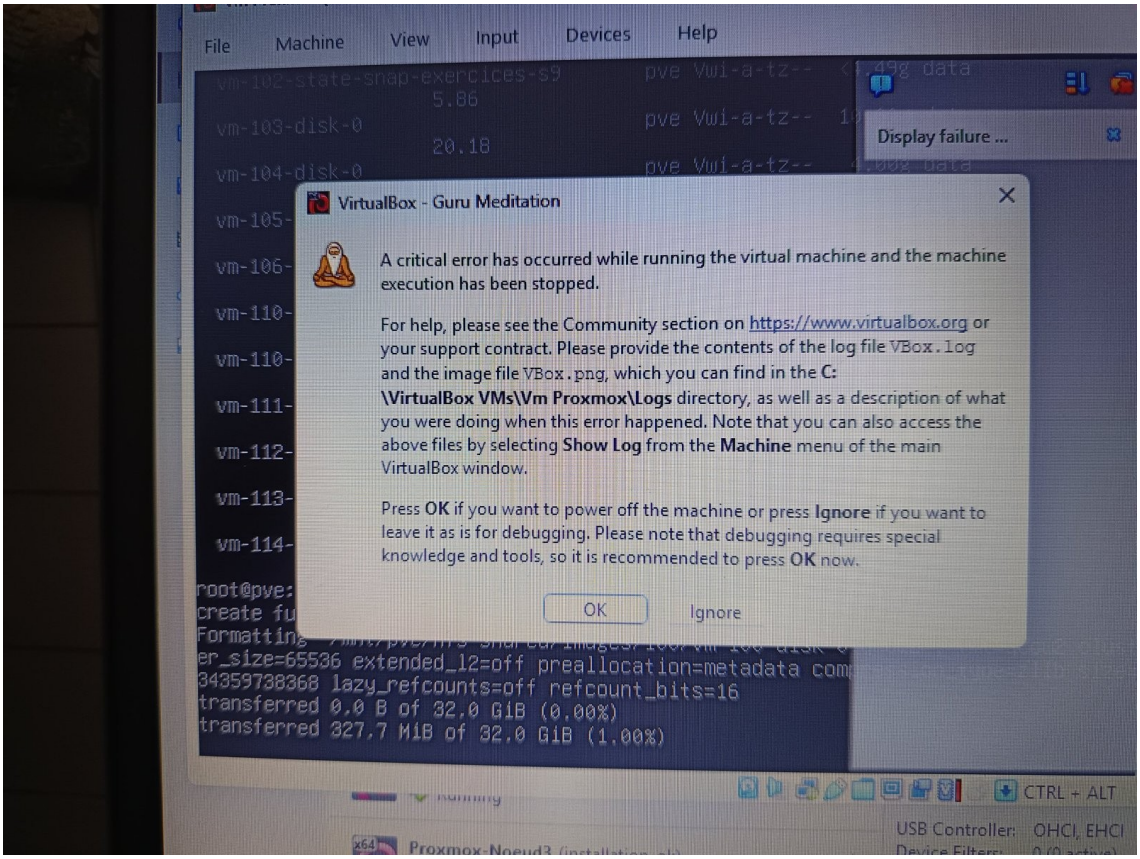
3.4 — VM-Test disk placed on nfs-shared

VM-Test is VM 100 (512 MB RAM, no ISO in the CD drive). Its 32 GB disk was on local-lvm.



qm list and qm config 100 before the operation: disk on local-lvm, 32 GB

Incident: moving the disk (qm disk move, equivalent to "Move Storage") crashed the pve VM (VirtualBox "Guru Meditation") at 1%, because the host PC ran out of resources. The original disk was untouched (deletion only happens after a successful copy). After restarting pve, we removed the incomplete copy with pvesm free.

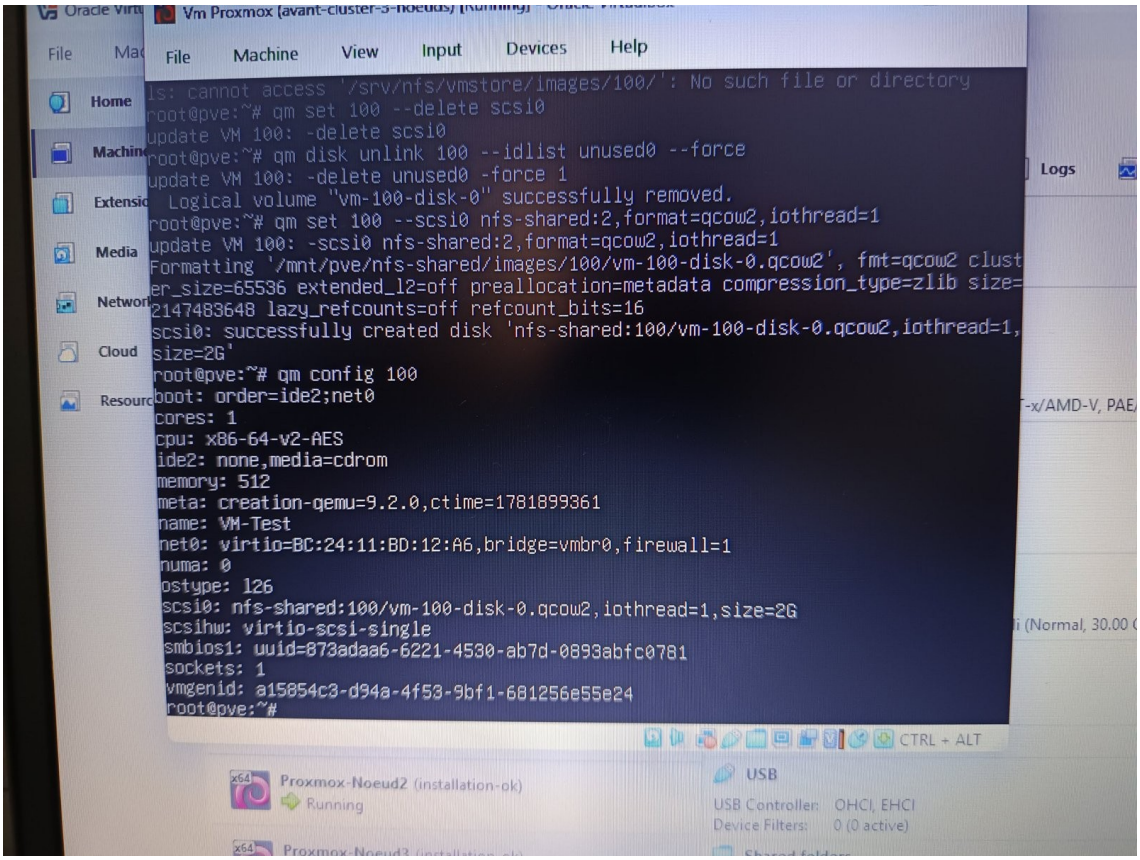


Crash of the pve VM during the disk copy (Guru Meditation)

The lvs command showed that VM-Test's disk was empty (Data% = 0.00). We therefore replaced it with a 2 GB disk created directly on nfs-shared: qm set 100 --delete scsi0, qm disk unlink 100 --idlist unused0 --force, then qm set 100 --scsi0 nfs-shared:2,format=qcow2,iothread=1. Same result for the rest of the lab: the VM's disk is on the shared storage.

```
  vm-100-disk-0                          pve Vwi-a-tz--  32.00g data
                   0.00
  vm-102-disk-0                          pve Vwi-a-tz--  10.00g data snap_vm-102-d
isk-0_snap-apache-ok 20.44
  vm-102-state-snap-apache-ok            pve Vwi-a-tz--  <4.49g data
                   11.80
  vm-102-state-snap-exercices-s9         pve Vwi-a-tz--  <4.49g data
                   5.86
  vm-103-disk-0                          pve Vwi-a-tz--  10.00g data
                   20.18
  vm-104-disk-0                          pve Vwi-a-tz--   4.00g data
                   3.63
  vm-105-disk-0                          pve Vwi-a-tz--   8.00g data
                   2.60
  vm-106-disk-0                          pve Vwi-a-tz--   8.00g data
                   2.60
  vm-110-disk-0                          pve Vwi-a-tz--   2.00g data
                   19.69
  vm-110-disk-1                          pve Vwi-a-tz--   2.00g data
                   32.53
  vm-111-disk-0                          pve Vwi-a-tz--   2.00g data
                   29.52
  vm-112-disk-0                          pve Vwi-a-tz--   2.00g data
                   32.57
  vm-113-disk-0                          pve Vwi-a-tz--   2.00g data
                   32.58
  vm-114-disk-0                          pve Vwi-a-tz--   2.00g data
                   32.53
root@pve:~# _
```

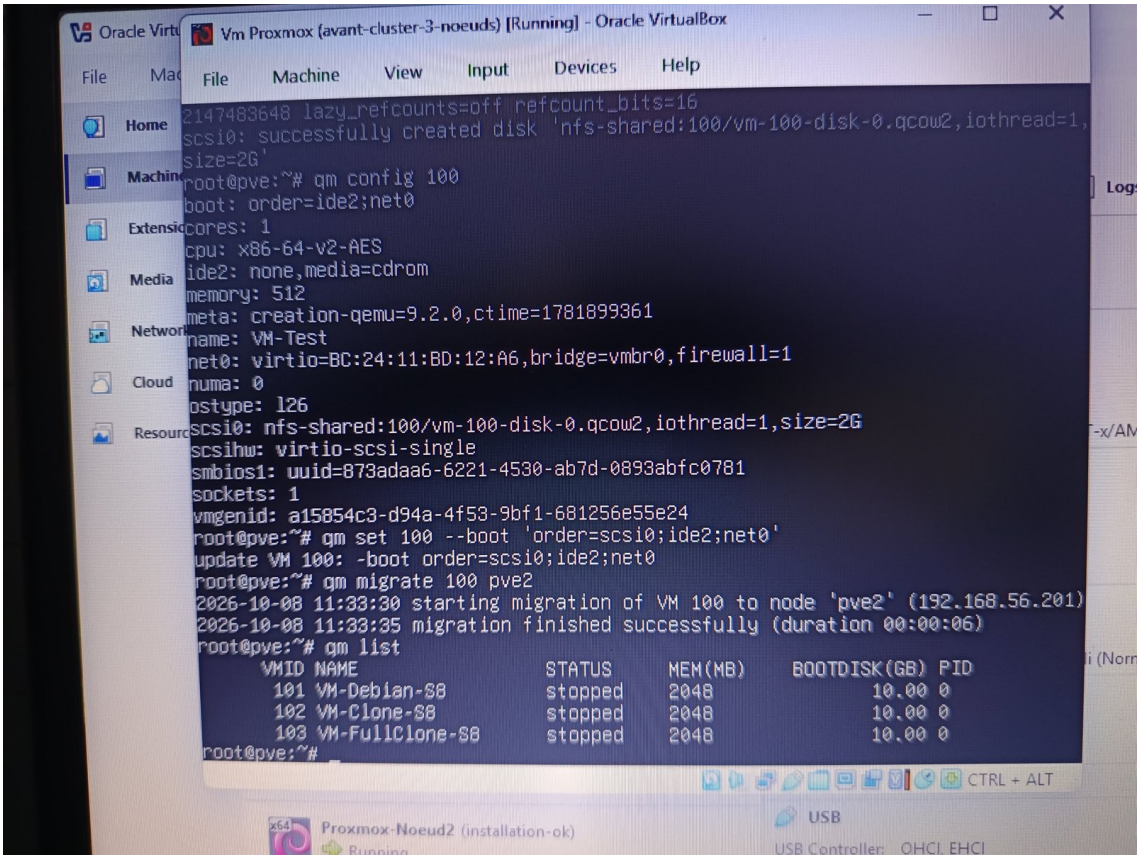
lvs: vm-100-disk-0 at 0.00% (empty disk)



qm config 100: scsi0: nfs-shared:100/vm-100-disk-0.qcow2, size=2G

3.5 — Migrating VM-Test to pve2

Boot order restored (qm set 100 --boot 'order=scsi0;ide2;net0'), then qm migrate 100 pve2. The migration took 6 seconds: only the VM's configuration moves; the disk stays on nfs-shared and is not copied.

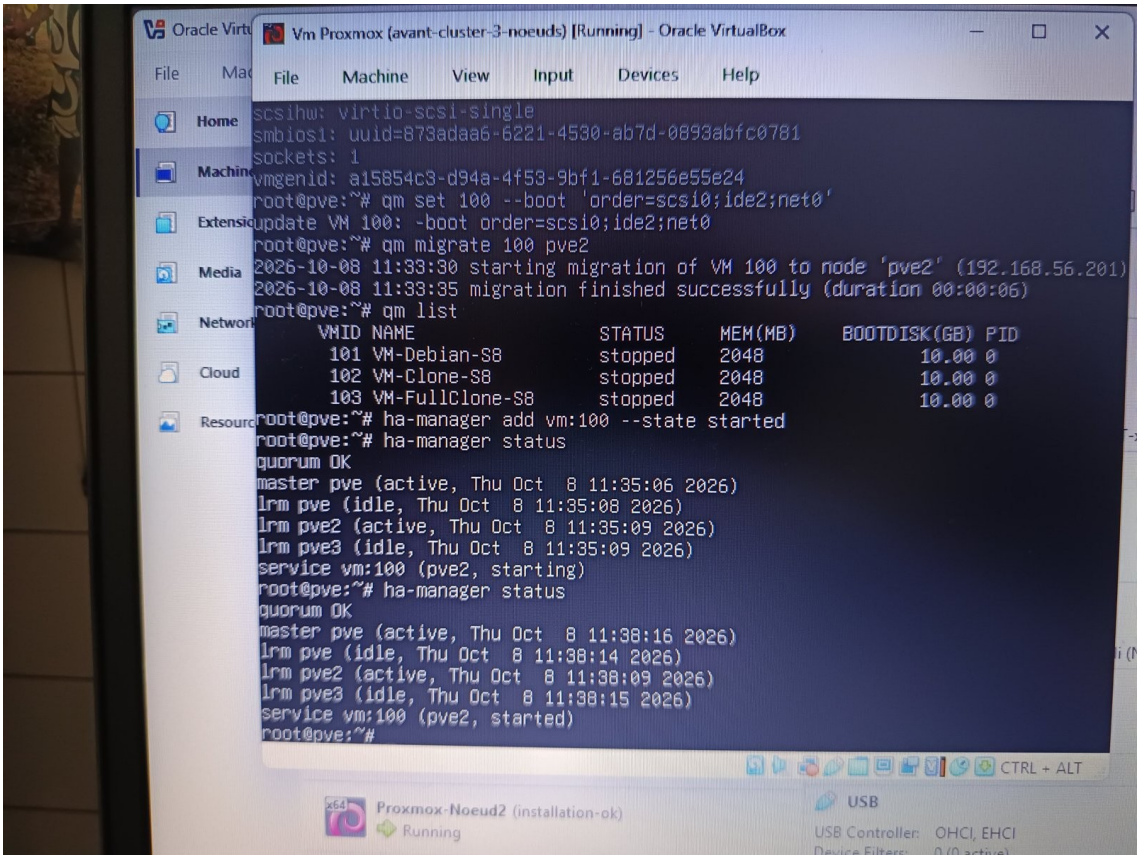


migration finished successfully (duration 00:00:06); VM 100 no longer listed on pve

PART 4 — High availability and failure test

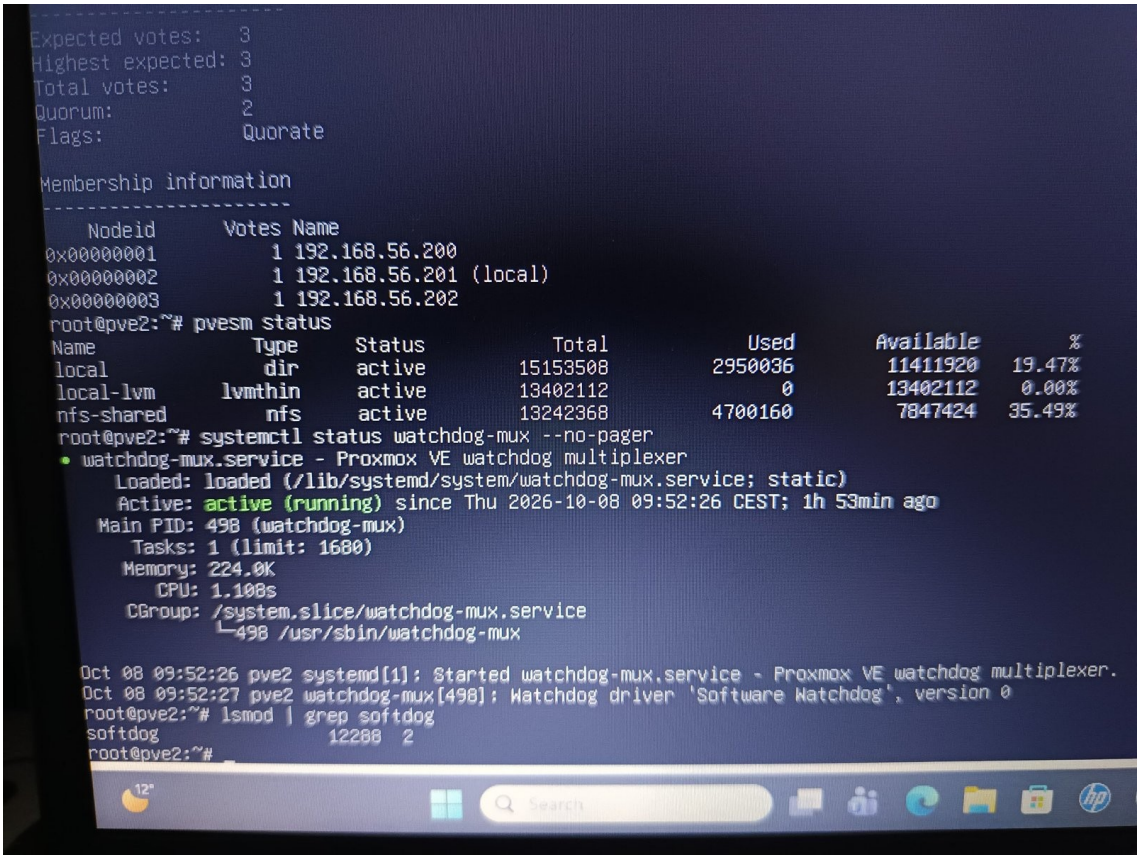
4.1 — Enabling HA on VM-Test

Command: ha-manager add vm:100 --state started. HA starts the VM on pve2.



ha-manager status: quorum OK, service vm:100 (pve2, started)

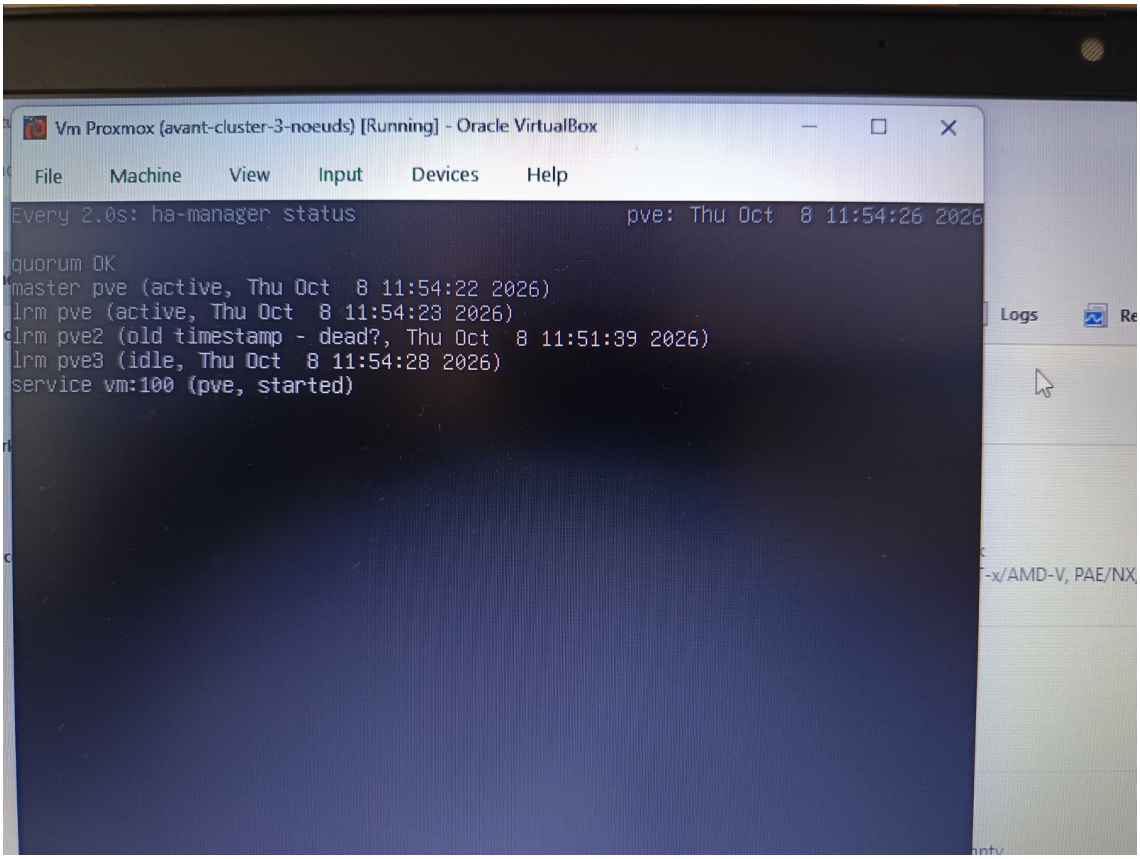
Monitoring mechanism on the host node (pve2): watchdog-mux is active and uses the "Software Watchdog"; the softdog module is loaded. If a node freezes, the watchdog reboots it, so it cannot keep running the VM while the VM is restarted elsewhere.



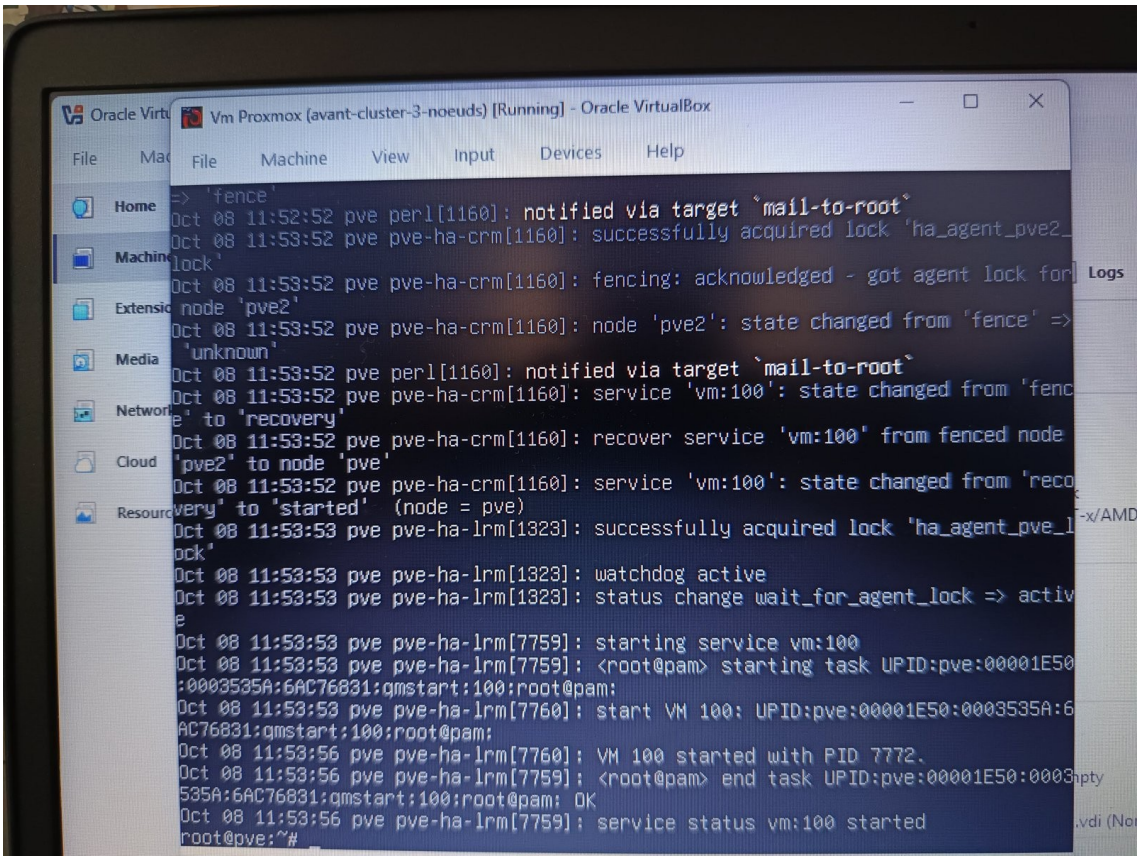
pve2: watchdog-mux active (running), Software Watchdog, lsmod | grep softdog

4.2 — Failure test and RTO measurement

Hard power-off of pve2 in VirtualBox (Machine > Close > Power off the machine). Times were read from the HA logs (journalctl -u pve-ha-crm -u pve-ha-lrm).



ha-manager status after the failure: pve2 "old timestamp - dead?", vm:100 (pve, started)



HA logs: fencing of pve2, recover service vm:100 to pve, VM 100 started

Measurement	Value
t0: node stopped (last signal from pve2)	11:51:39
t1: service restarted (ha-manager)	11:53:52
t2: VM usable again (VM 100 started)	11:53:56
Measured RTO (t2 minus t0)	2 min 17 s
Calculation	11:53:56 − 11:51:39 = 2 min 17 s (137 s)
Target RTO (set by the group, see below)	15 min
Gap	15 min − 2 min 17 s = 12 min 43 s of margin: COMPLIANT

Evidence: t0 is readable on the ha-manager status screenshot ("lrm pve2 (old timestamp - dead?, … 11:51:39)"); t1 and t2 are readable on the HA log screenshot ("recover service 'vm:100' … 11:53:52" and "VM 100 started … 11:53:56"). Calculation checked by hand: from 11:51:39 to 11:53:39 = 2 min, plus 17 s = 2 min 17 s.

Requirements-spec target: our spec does not set an RTO ("to be confirmed with the client"), so the group sets it at 15 min for a node failure. Justification: online booking is Santélia's critical service; the March outage blocked it for over 6 h; a 15-minute interruption is tolerable for patients and reception (phone booking meanwhile), and HA restarts the VM without any action from the 2-person IT team.

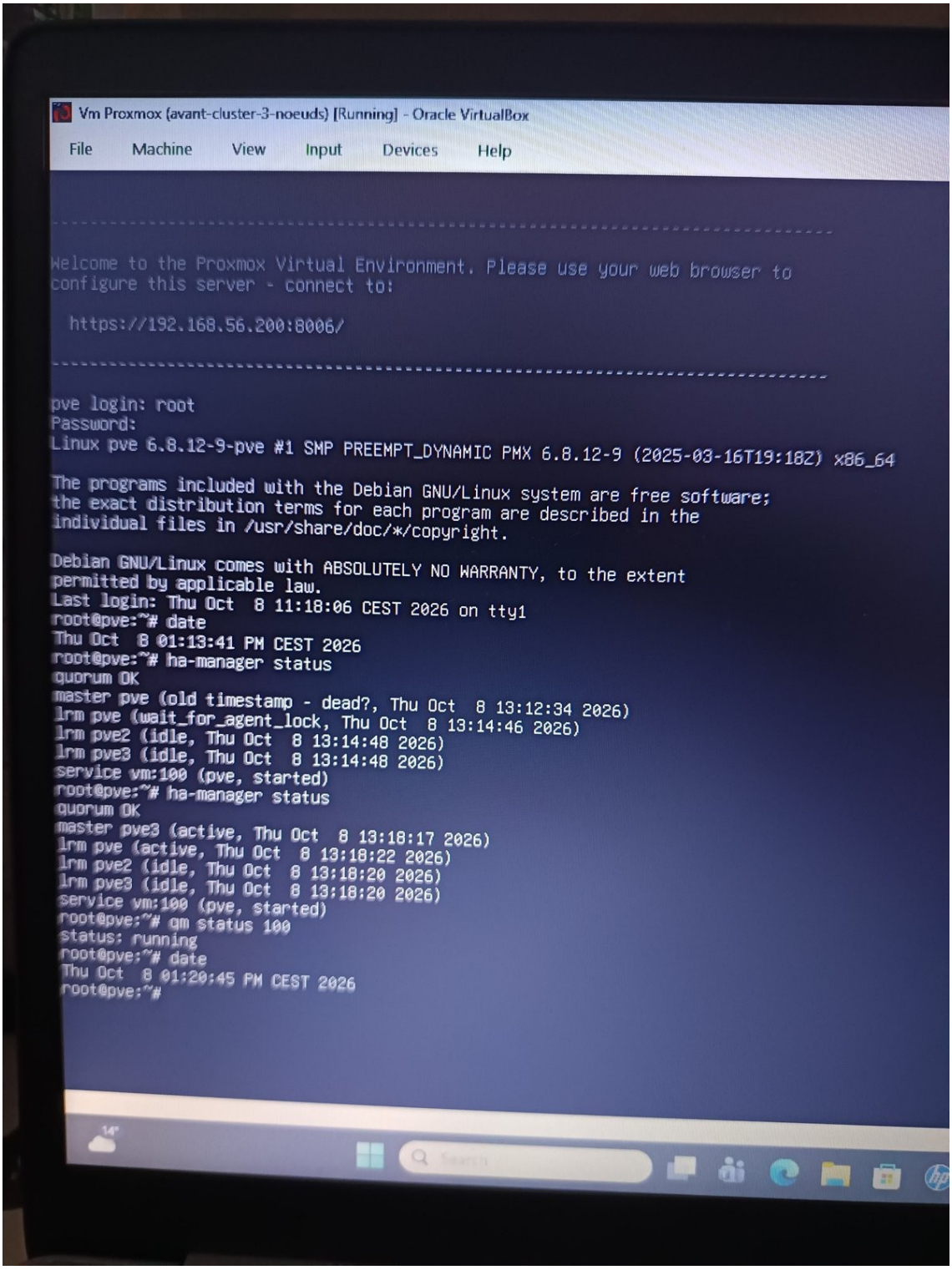
Analysis: almost all of the delay (about 2 min 13 s) is the safety wait during which Proxmox makes sure pve2 is really stopped (fencing) before restarting the VM. The restart itself took 4 seconds. HA chose to restart the VM on pve (not on pve3): HA picks the target node among the available nodes.

Limits: t0 is pve2's last signal, a few seconds before the actual power-off. VM-Test has no operating system: "usable" here means "VM started". With a real system, its boot time would have to be added (roughly 30 to 60 s).

Restoring normal state

pve2 was powered back on and rejoined the cluster (Nodes: 3, Quorate: Yes). The VM stayed on pve: Proxmox does not automatically move a VM back to its original node (no "failback"), to avoid a second interruption.

Incident: after the host PC went to sleep, the clocks of pve and pve3 were about 36 minutes behind (HA reported "detected time drift"). The sudden time correction (chronyc makestep) caused pve to be rebooted automatically by its watchdog (HA self-protection). HA then put everything back in order on its own: the master role moved to pve3 and VM-Test was running again. Lesson: time synchronisation (NTP) is essential in a cluster.



Final state: quorum OK, master pve3, vm:100 (pve, started), qm status 100: running

Comprehension questions

1. Why can't the VM restart elsewhere if its disk is stored only on the failed node?

A VM's disk is a file (or volume) holding its whole system and data. If it is stored on the failed node's local disk, the other nodes cannot access it: they can read the VM's configuration, but not its disk. Without a disk, the VM cannot start. That is why we put the disk on shared storage (NFS), accessible by all 3 nodes. Another solution is ZFS replication, which regularly copies the disk to the other nodes.

2. What is the difference between a BCP (PCA) and a DRP (PRA) in the Santélia case?

The BCP (Business Continuity Plan, PCA) keeps the service running during a failure, with no interruption or a very short one. For Santélia: if a server fails, the cluster automatically restarts the appointment or patient-records software on another server within minutes (2 min 17 s in our test).

The DRP (Disaster Recovery Plan, PRA) restarts the service after a more serious disaster, when the whole site or the data is affected: fire in the server room, ransomware. Data is restored from backups (Proxmox Backup Server, off-site copy, 3-2-1 rule). Recovery takes longer (several hours) and data since the last backup may be lost (RPO).

3. Does your measured RTO meet your requirements spec? If not, what should change: the technical solution or the requirement?

Our requirements spec did not set an RTO; the group set the target at 15 min for a node failure (justified above, Part 4.2). Measured RTO: 2 min 17 s, i.e. 12 min 43 s of margin: the target is met and nothing needs to change. If management required less than one minute of downtime, the technical solution would have to change (active-active applications, load balancing), because HA's safety wait (fencing) alone takes about 2 minutes.

Improvement point: in our lab, the NFS share is hosted on pve. If pve fails, the shared storage disappears: it is a single point of failure. In production, the shared storage itself would need to be redundant (NAS in high availability, or Ceph), and the client should approve numeric RTO and RPO targets.

Lab HA/BCP — Session B: back up, restore, write the recovery plan

GRETA de l'Essonne — AIS professional title — Block 2: Design and evolution

Trainer: Samir KECHEMIR

Charles Maduba — Group 3 (work carried out in a team of 4 trainees)

Session B: Monday 12 October 2026. Follows session A (cluster, NFS, HA, RTO = 2 min 17 s).

English version of the lab report, for personal reference. The original report is written in French.

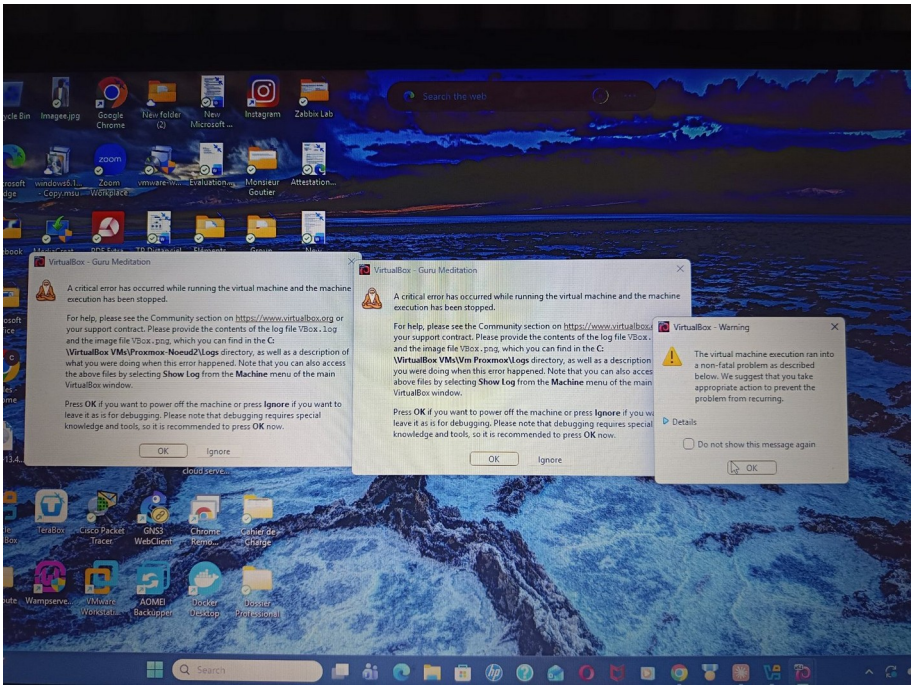
Summary

Part	Content	Result
0	Catch-up: cluster, HA, shared disk	Checked
1	Backup, restore, automatic backup	Backup 10 s, restore 3.6 s, daily job at 21:00 (keep 7)
2	RPO	Estimated per scenario (VM without an OS)
3	Disaster recovery plan (DRP)	Written (6 sections)
4	Cross-group presentation	Group 3 values ready

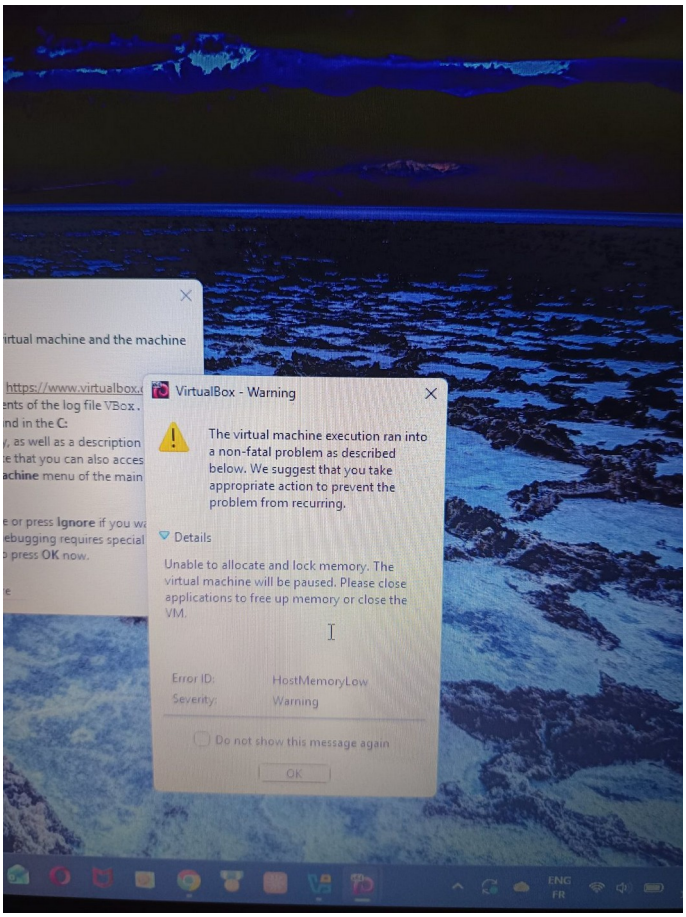
All operations were done on the command line (vzdump, qmrestore, pvesh): with the 3 nodes running, the host PC does not have enough memory to open the web interface. These commands are the documented equivalents of the web interface buttons (the lab guide lists them).

Start-up incident: not enough memory on the host PC

On the first start of the 3 nodes, VirtualBox stopped pve and pve2 ("Guru Meditation") with the error HostMemoryLow: "Unable to allocate and lock memory". The 3 nodes need 6 GB of RAM on an 11.8 GB PC where Chrome, the Claude app and WhatsApp were also running. After closing these applications, the 3 nodes started normally.



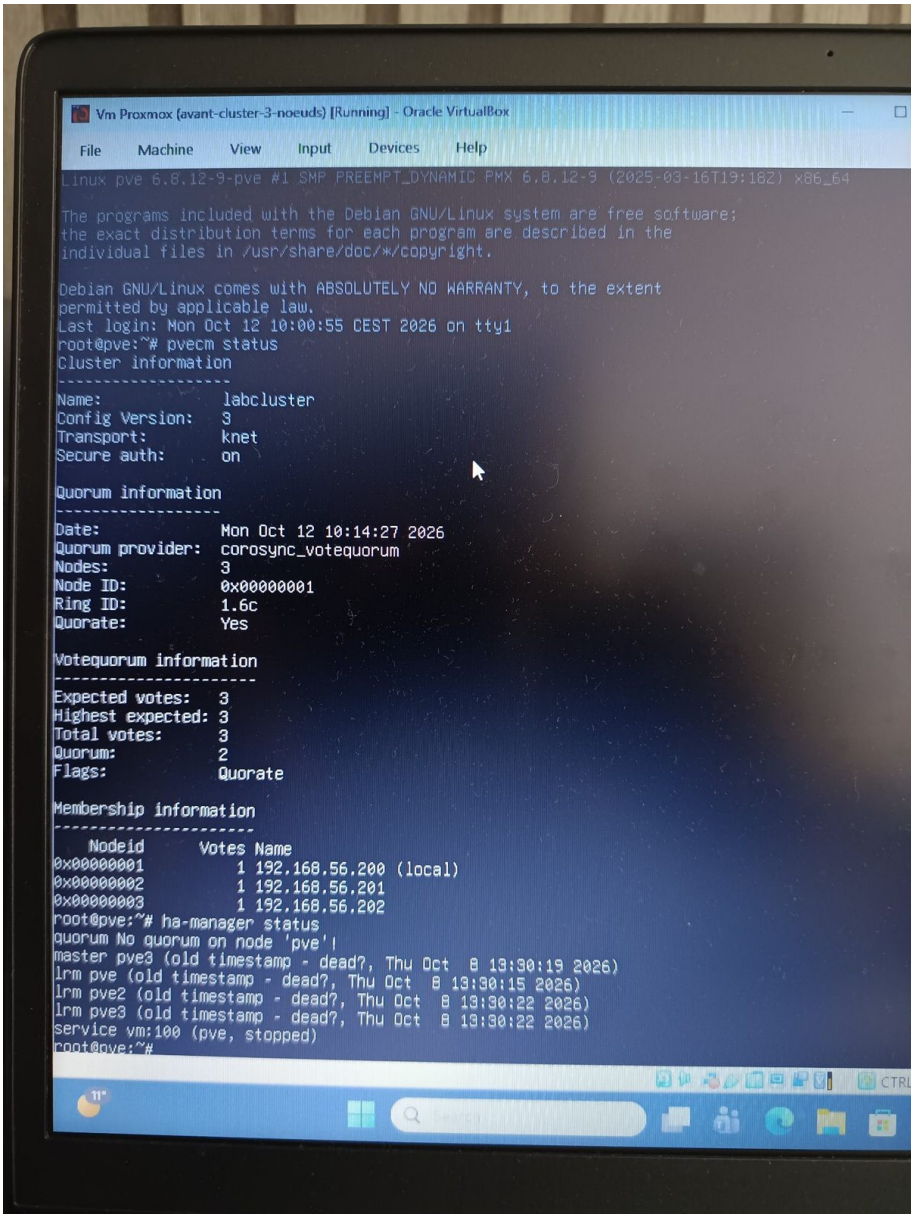
pve and pve2 stopped by VirtualBox (Guru Meditation)



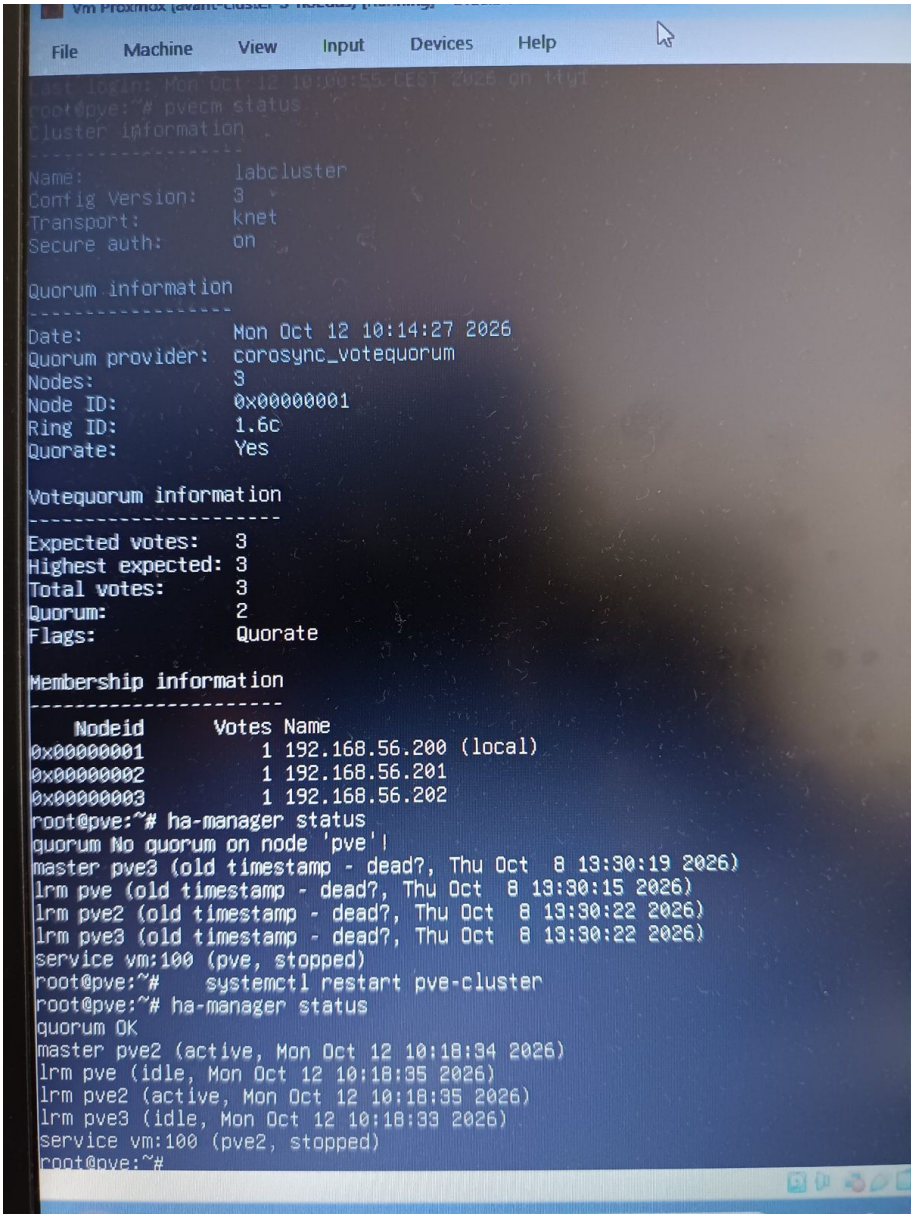
Error detail: HostMemoryLow

PART 0 — Checking session A

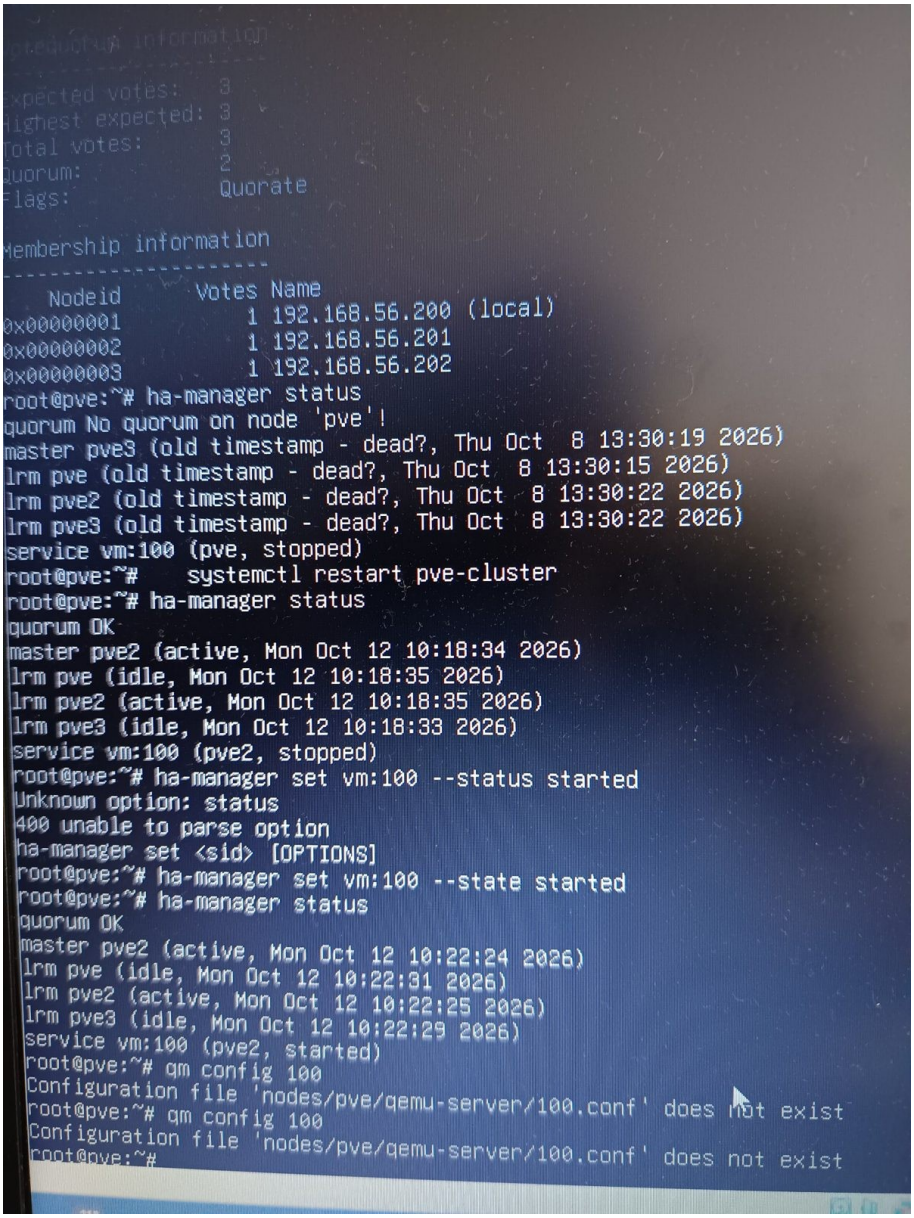
pvecm status showed 3 votes and Quorate: Yes, but ha-manager status showed "No quorum on node 'pve'" with timestamps from 8 October: the cluster configuration service (pmxcfs) had stayed stuck at start-up. Fixed with systemctl restart pve-cluster (same problem as in session A). HA then took over again and placed VM-Test on pve2.



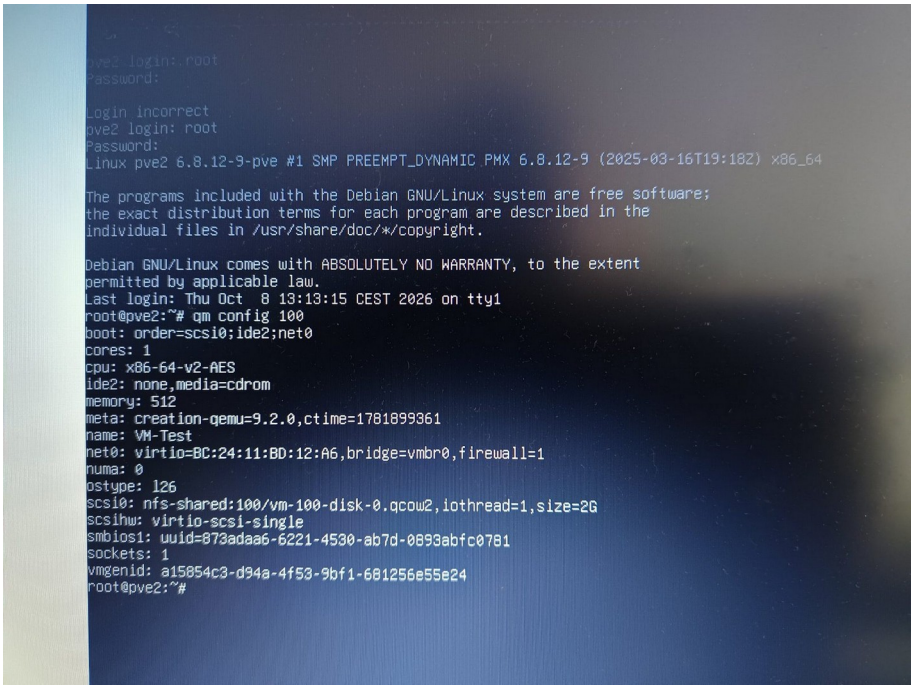
pvecm status: 3 votes, Quorate: Yes; ha-manager: timestamps from 8 October



After systemctl restart pve-cluster: quorum OK, VM 100 on pve2



ha-manager set vm:100 --state started → service vm:100 (pve2, started)

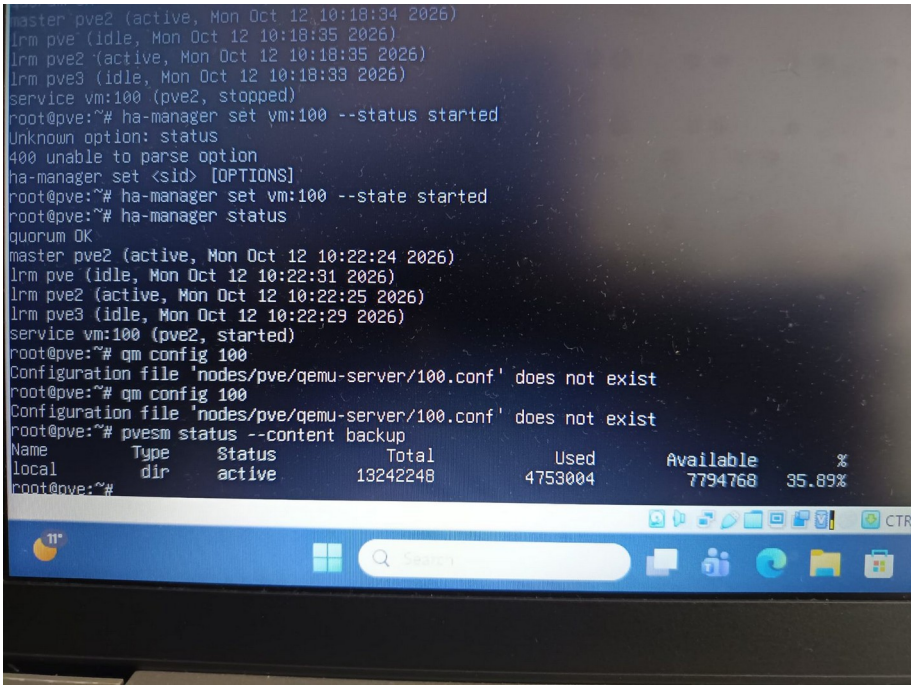


qm config 100 on pve2: scsi0 disk on nfs-shared, 2 GB

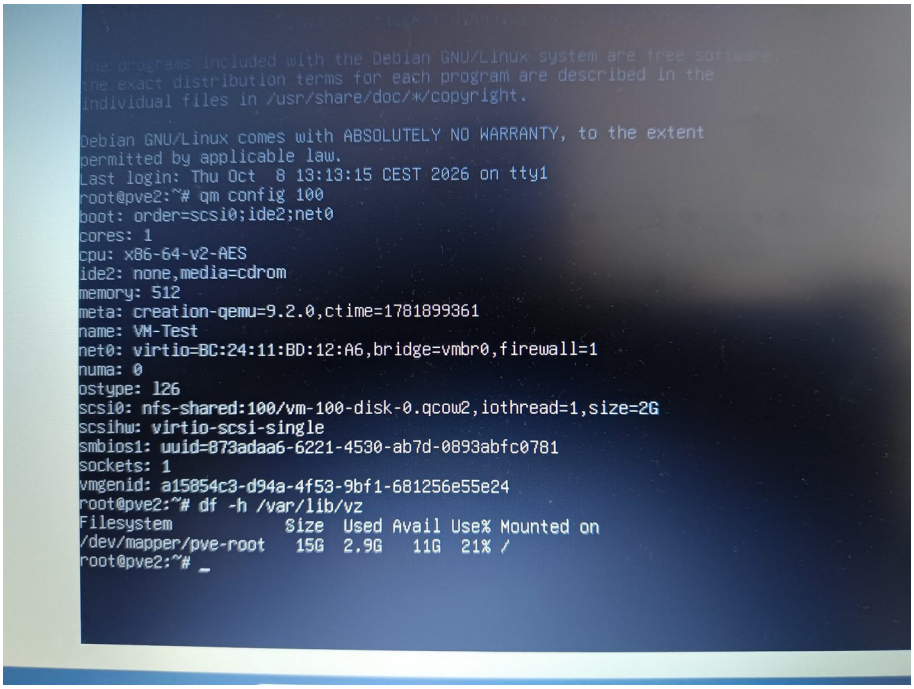
PART 1 — Back up and restore

1.1 — Where to store the backups

The local storage (/var/lib/vz) accepts backups. On pve2 (the node running the VM), 11 GB are free, far more than VM-Test's 2 GB disk.



pvesm status --content backup: local, dir, active

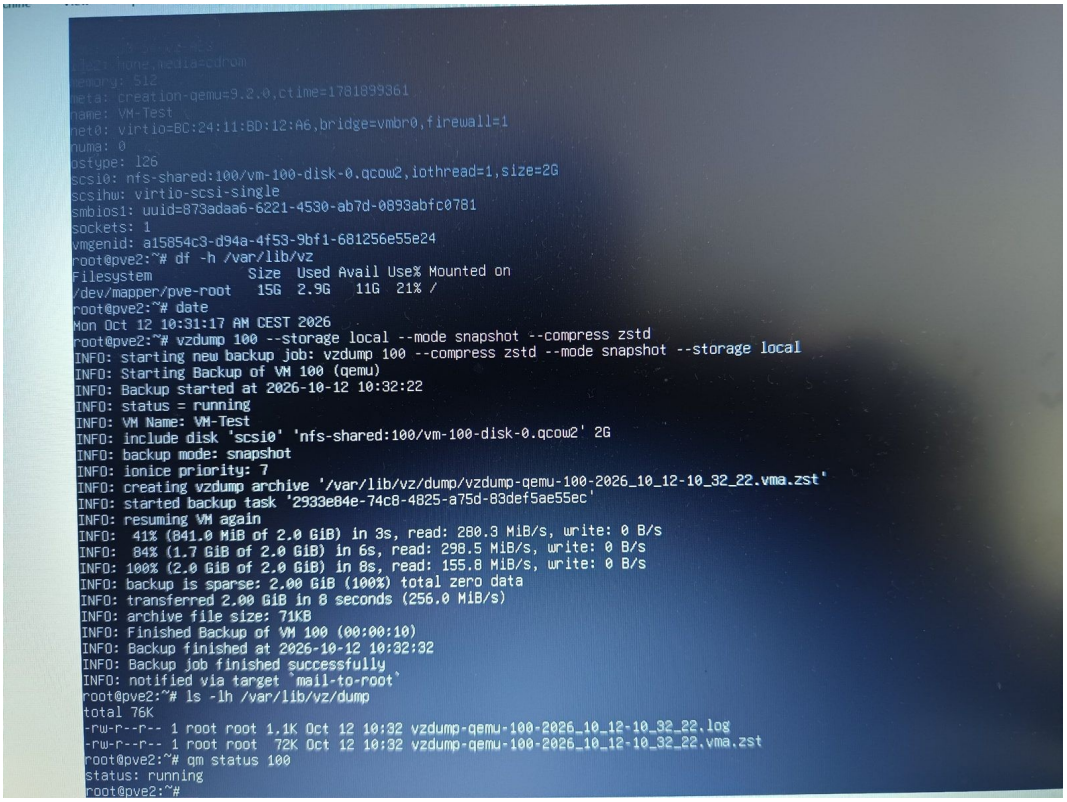


df -h /var/lib/vz on pve2: 11 GB available

1.2 — Manual backup

```
vzdump 100 --storage local --mode snapshot --compress zstd
```

Measurement	Value
Start / end	10:32:22 / 10:32:32
Duration	10 seconds
Archive size	72 KB (2 GB disk entirely empty: "sparse 100%")
Does the VM keep running?	Yes: snapshot mode, qm status 100 → running



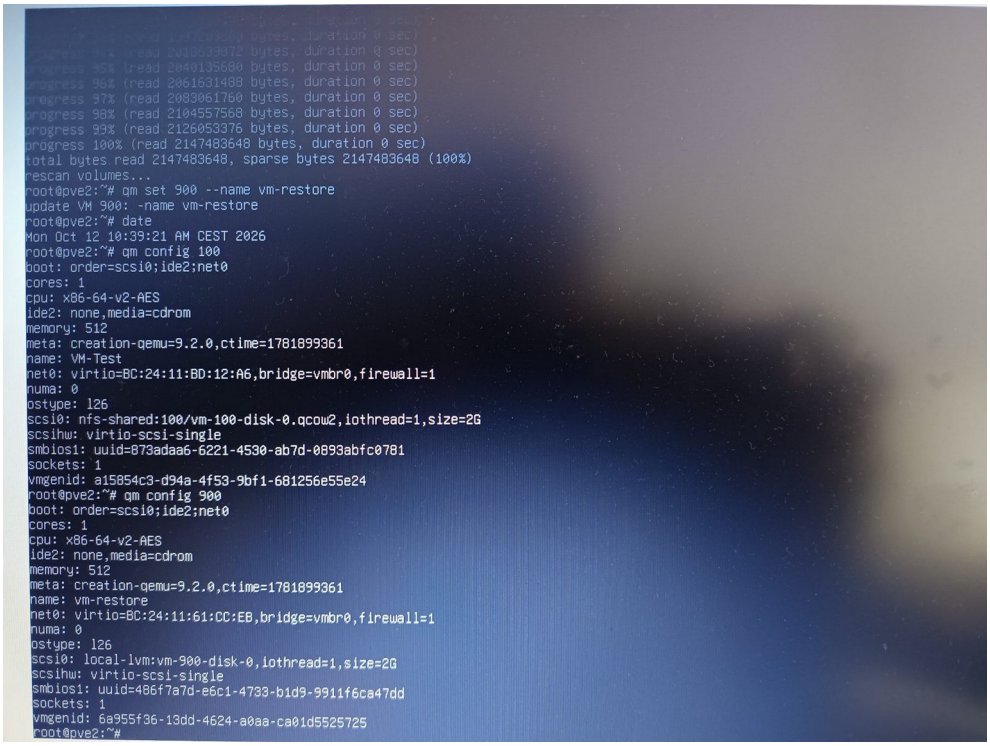
vzdump backup finished successfully, archive in /var/lib/vz/dump, VM still running

1.3 — Test restore as ID 900

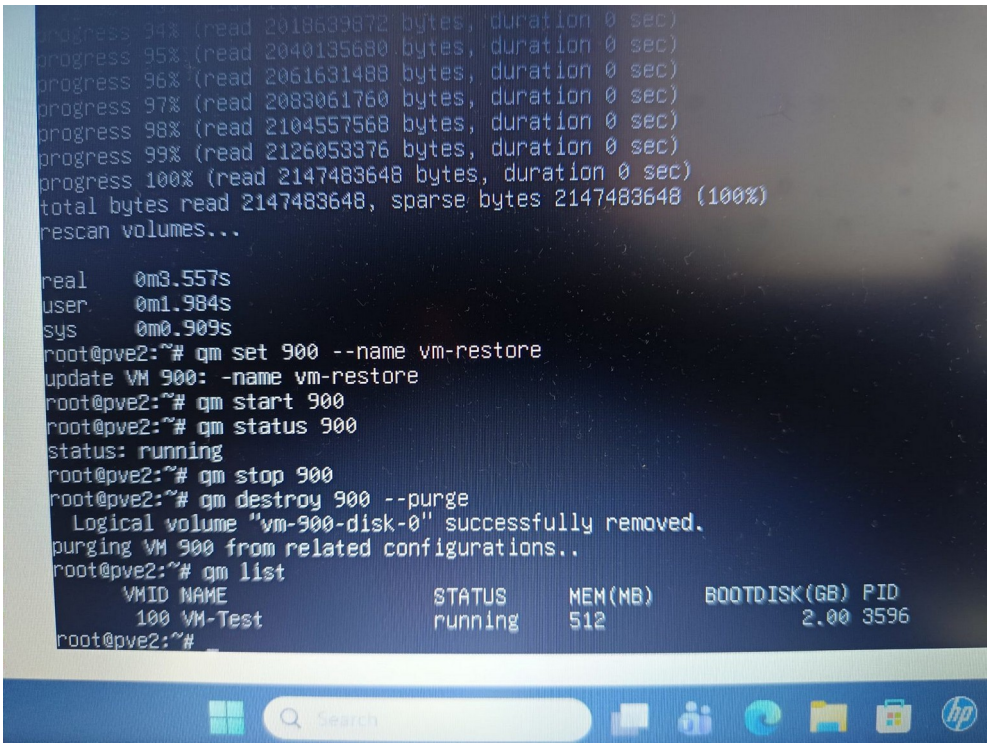
`qmrestore /var/lib/vz/dump/vzdump-qemu-100-2026_10_12-10_32_22.vma.zst 900 --storage local-lvm --unique 1`

The restore uses another ID (900), on local-lvm, with --unique (new MAC address), so as not to overwrite the running VM 100 or disturb HA. The first restore succeeded but its start time had scrolled off the screen; it was redone with the time command, which measures the exact duration.

	VM 100 (VM-Test)	**VM 900 (vm-restore)**
RAM / CPU	512 Mo / 1	identical
Disk	nfs-shared, 2 Go	local-lvm, 2 Go
MAC address	BC:24:11:BD:12:A6	BC:24:11:61:CC:EB (different)
Restore duration	—	3.6 s (real 0m3.557s)



Comparing qm config 100 / qm config 900



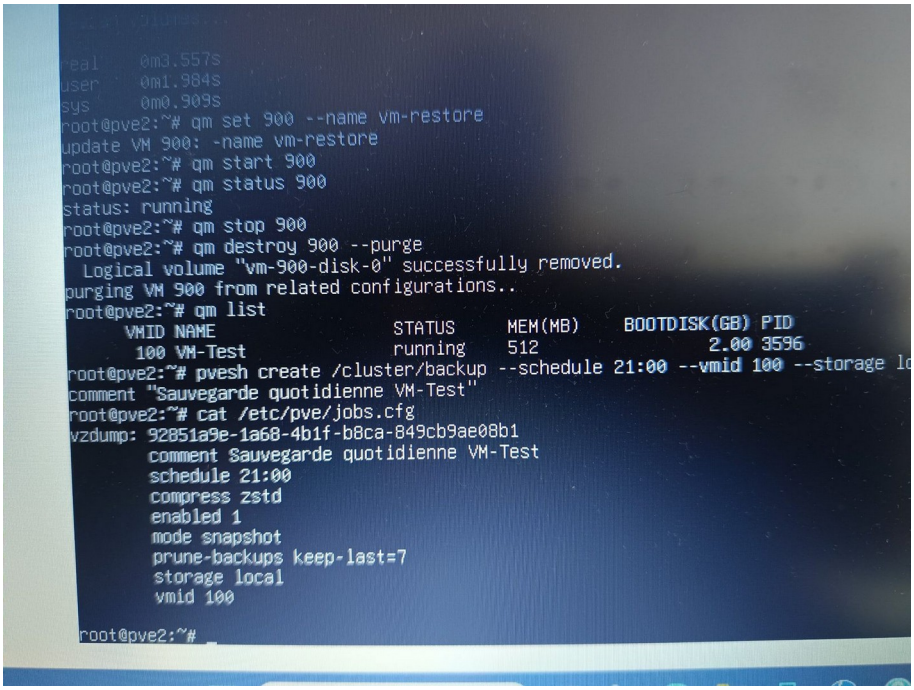
Timed restore (3.6 s), VM 900 started (running) then deleted; only VM 100 remains

VM-Test has no operating system: proof of start-up is the "running" state (the console would only show the BIOS). The 3.6 s duration is not representative of a real VM: a full 50 GB disk would take several minutes.

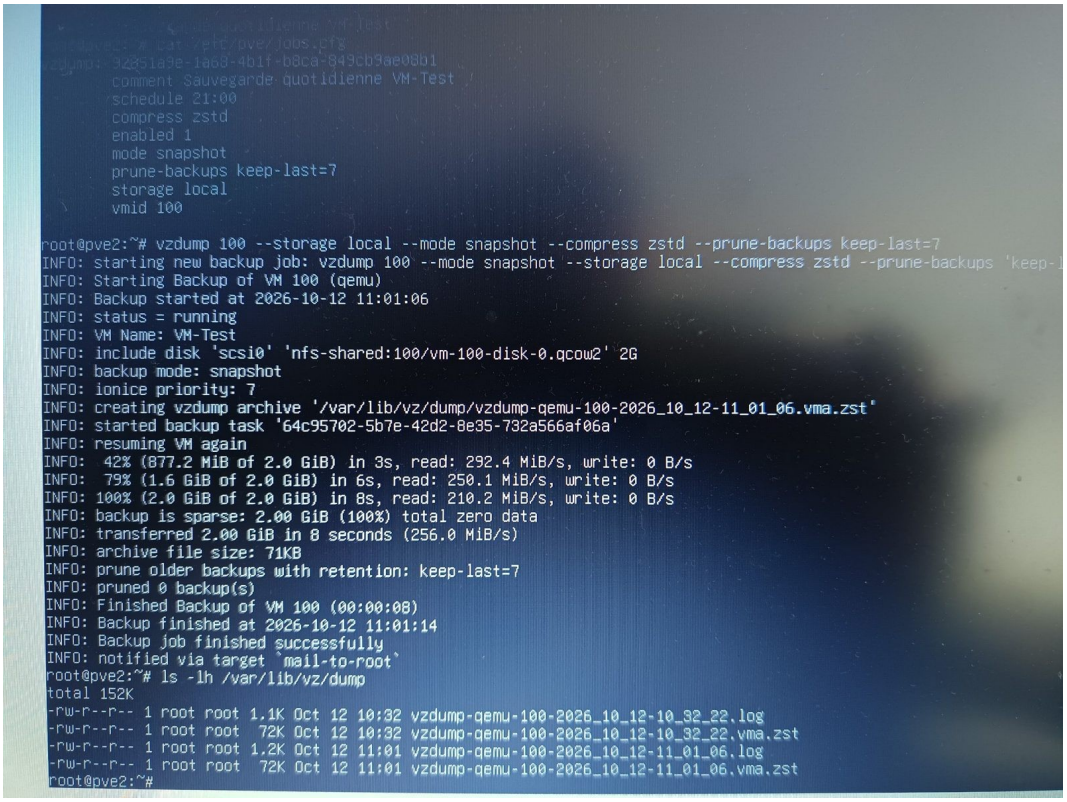
1.4 — Automatic backup

```
pvesh create /cluster/backup --schedule 21:00 --vmid 100 --storage local --mode snapshot --compress zstd --prune-backups keep-last=7 --enabled 1
```

Scheduled job every day at 21:00, snapshot mode, keeping 7 backups (older ones are deleted automatically). "Run now" was done by running vzdump with the same settings: backup in 8 s, "pruned 0 backup(s)" (nothing to delete, only 2 backups).



/etc/pve/jobs.cfg: schedule 21:00, prune-backups keep-last=7



"Run now": new backup at 11:01, 2 backups in local storage

1.5 — Does the lab follow the 3-2-1 rule?

Question	Answer
How many copies of the disk? Where?	3: the original on nfs-shared (pve's disk) and 2 backups in /var/lib/vz/dump on pve2
On how many physical media?	1 in reality: the 3 nodes are VMs on the same PC disk
An off-site copy?	No
If the PC burns, what is left?	Nothing
What would Santélia need?	A Proxmox Backup Server on separate hardware (2nd medium) and an off-site copy, ideally immutable (against ransomware), as planned in the requirements spec

Conclusion: 3 copies, but neither 2 media nor an off-site copy. The lab therefore does not follow the 3-2-1 rule.

PART 2 — RPO: how much data do we lose?

2.1 — RPO by scenario (backup every evening at 21:00)

Scenario	What saves you?	RPO (maximum loss)
The node running the VM fails	HA (disk on NFS shared storage)	≈ 0 (only data in memory not yet written)
VM deleted by mistake at 18:00	Last night's 21:00 backup (HA cannot help)	≈ 21 h
Ransomware encrypts the VM's data	The last clean backup, from before the infection	≥ 24 h, up to 7 days (retention)
pve1 (NFS server) destroyed	The backup stored on pve2 (HA cannot help: the shared disk is gone)	Up to 24 h

ZFS groups: replication every 15 min	ZFS replication + HA (not applicable: our group uses NFS)	Up to 15 min

HA protects against a node's hardware failure, not against deletion, ransomware or loss of the storage: in those cases only a backup lets you go back in time. Lab limit: the backups are on the same PC as the data.

2.2 — Practical measurement

Not carried out: VM-Test has no Linux system. The RPO was estimated (table 2.1), not measured, as the lab guide allows.

PART 3 — Santélia disaster recovery plan (DRP)

The document to open on the day of a disaster. It must let someone who did not build the infrastructure bring the service back.

1. Scope

- Services covered: online appointment booking for the 3 clinics (Évry, Corbeil-Essonnes, Ris-Orangis) and the patient records software, identified as critical in the requirements spec.
- Infrastructure: 3-node Proxmox cluster "labcluster" (pve, pve2, pve3), critical VM disks on nfs-shared shared storage, VMs protected by HA.
- Backups: every day at 21:00, 7 backups kept. In the lab, the appointment VM is represented by VM-Test (VM 100).
- Out of scope: workstations, telephony, medical equipment.

2. Disaster scenarios

Scenario	Solution	RTO	RPO	BCP or DRP?
Node failure	HA + NFS shared storage	2 min 17 s (measured)	≈ 0	BCP (PCA)
Storage server failure	Restore from backup on another node (local-lvm)	3.6 s measured (empty disk); estimated 15–60 min in production	≤ 24 h	DRP (PRA)
Accidental VM deletion	Restore from the latest backup	Same	≤ 24 h	DRP (PRA)
Ransomware	Isolate, then restore the last clean backup (ideally an immutable off-site copy)	Several hours (analysis + restore)	≥ 24 h, up to 7 days	DRP (PRA)
Fire in the Évry server room	Rebuild on another site from the off-site copy (to be set up)	Hours to days	≤ 24 h with a daily off-site copy	DRP (PRA)

3. Step-by-step restore procedure

1. Confirm the incident and alert the person who decides to trigger the DRP (section 4).
2. Log in as root on a working node and check the cluster: pvecm status → Quorate: Yes.
3. Find the backup: ls -lh /var/lib/vz/dump/ on the node that ran the VM (web interface: node > local > Backups). Pick the most recent vzdump-qemu-<id>-<date>.vma.zst from before the incident (ransomware: from before the infection).

4. Restore under a new ID, without overwriting the old VM: qmrestore <file> <new id> --storage local-lvm --unique 1 (web interface: Restore, Storage local-lvm, tick Unique).
5. Check the configuration: qm config <id> (RAM, CPU, disk).
6. Start it: qm start <id>, then qm status <id> → running. Open the console.
7. Check the service: open the appointment page, log in with a test account, create then cancel a test appointment.
8. Put the VM back under HA: ha-manager add vm:<id> --state started (and remove the old HA resource).
9. Announce that the service is back. Record start and end times (actual RTO) and the period of lost data (actual RPO) in the incident log.

4. Roles and responsibilities

Role	Who?	What they do
Decides to trigger the DRP	Santélia management (delegated to IT outside office hours)	Assesses the impact, approves the trigger and the accepted data loss
Carries out the restore	The IT team (2 people: a lead and a backup)	Follows the procedure, checks the service, records the times
Informs the clinics and patients	Communication / clinic reception lead, with management	Informs staff, organises phone booking as a fallback, contacts the patients affected

5. Targets compared with measurements

The requirements spec leaves RTO and RPO "to be confirmed with the client". As the guide requires, the group sets the targets and justifies them:

- RTO node failure: 15 min. Booking is the critical service; the March outage blocked it for over 6 h; 15 min is tolerable (phone booking meanwhile) and HA acts without human intervention.
- RTO restore: 4 h. A 2-person IT team must be able to restore within half a day, well under the 6 h of the March outage.
- RPO node failure: 0. The disk is on shared storage, no written data should be lost.
- RPO restore: 4 h. Lose at most half a day of appointments, which reception can rebuild by calling patients back.

Indicator	Target (set by the group)	Measured in the lab	Compliant?
RTO (node failure)	15 min	2 min 17 s (11:53:56 – 11:51:39)	Yes
RTO (restore)	4 h	3.6 s restore (empty disk, to re-measure with a real VM)	Yes in the lab
RPO (node failure)	0	≈ 0	Yes
RPO (restore)	4 h	Up to 24 h (backup at 21:00; VM deleted at 18:00 → 21 h lost)	NOT COMPLIANT

Proposed fix for the RPO: move from 1 backup a day to 1 backup every 4 h (snapshot mode, no service interruption). It is the backup frequency that must change, not the target: losing a whole day of appointments is not acceptable for Santélia.

6. Testing and updating the plan

- Daily: automatic check of backup results, with an alert on failure (Zabbix or email).
- Monthly: test restore under another ID, as in Part 1.3, measuring the duration.

- Yearly, and before the HAS audit: full disaster exercise (simulated failure and restore).
- Updated by IT after every test, every real incident and every infrastructure change (new server, new VM, new provider), and at least once a year. Each version is dated and approved by management.

PART 4 — Cross-group presentation

	Group 1	**Group 2**	**Group 3**
Storage (NFS / ZFS)	Replicated ZFS	Replicated ZFS	NFS
Measured RTO	Between 2 min and 3 min 30 (class)	Between 2 min and 3 min 30 (class)	2 min 17 s (node failure)
Measured or estimated RPO	Up to the replication interval (5–15 min)	Up to the replication interval (5–15 min)	≈ 0 (HA); ≤ 24 h (backup)
Single point of failure	No	No	Yes: the NFS share hosted on pve (and the single PC)

Groups 1 and 2 values taken from the class summary presented by the trainer.

Comprehension questions

1. HA is in place: why are backups still needed?

HA only protects against a node failure: it restarts the VM elsewhere, in the state it was in. It does not protect against a deleted VM, data encrypted by ransomware (the VM would restart with its encrypted data), corruption, loss of the NFS storage or a site disaster. Only a dated copy lets you go back in time.

2. Why restore the test backup under a different VM number?

So as not to overwrite VM 100, which is running and protected by HA, and to test the backup without interrupting the service. The Unique option gives a new MAC address to avoid a network conflict. If the backup is bad, the original stays intact.

3. Your backup is on the local storage of the node running the VM. What happens if that node is destroyed?

The backups disappear with the node. In our lab, if pve2 is destroyed, HA restarts VM-Test elsewhere (its disk is on NFS), but all the backups are lost: after a later ransomware attack or deletion, there would be nothing left to restore. A backup must live on separate storage (Proxmox Backup Server on another machine) with an off-site copy: the 3-2-1 rule.

4. For Santélia's appointment booking, is a low RPO or a low RTO more important? Explain.

A low RTO comes first: while the service is down, patients cannot book and the clinics' activity is blocked (the March outage lasted over 6 hours). Losing a few recent appointments can be recovered (calling patients back, confirmations sent by email or SMS), so an RPO of a few hours is acceptable. For patient records it is the opposite: medical data must not be lost, so a low RPO comes first.

Takeaways and lessons

- A backup is only worth something if its restore has been tested: here, a 10 s backup and a proven 3.6 s restore.

- HA (BCP) and backups (DRP) complement each other: one keeps the service running, the other lets you go back in time.
- The lab does not follow the 3-2-1 rule: everything is on one PC. In production, a separate backup server and an off-site copy are needed.
- Limited resources: close unneeded applications before starting the 3 nodes, and check ha-manager status after every start-up.